VCS JOINT VALIDATION & VERIFICATION REPORT TEMPLATE

KAMPONG CHHNANG 60 MW SOLAR POWER PROJECT

Document Prepared by:

4K Earth Science Private Limited

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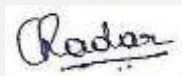
Client	Prime Road Alternative (Cambodia) Co., Ltd.
Prepared by	4K Earth Science Private Limited.
Approved by	 Chandrakala R Managing Director
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Summary:

Brief summary on the validation and verification and the project activity

'4K Earth Science Private Limited' has been contracted by 'Kosher Climate India Private Limited' to perform the Combined validation and verification of the first monitoring period of the VCS project activity "Kampong Chhnang 60 MW Solar Power Project", with regard to the relevant requirements of VCS Standard version 4.7/6/.

Project: The project activity "Kampong Chhnang 60 MW Solar Power Project" is a project which employs the approved CDM large-scale methodology ACM0002: "Grid-connected electricity generation from renewable sources", Version 21.0/9/.

In 2019, ADB's Private Sector Operations Department (PSOD) facilitated the bidding process by issuing letters of support and providing indicative term sheets to 20 of the 26 bidders, enabling multiple bidders to incorporate identical financing terms, enhancing competitive fairness and ensuring that the advantages of concessional finance were reflected in the final tariff bid prices. Prime Road Alternative (Cambodia) Company Limited was awarded the project through this reverse auction tender process, resulting in a record-low utility-scale solar tariff of $0.03877 per kWh in Southeast Asia. Additionally, the project proponent has entered into long-term Power Purchase Agreements (PPA) with Electricite du Cambodge (EDC) to provide the generated power at the agreed-upon price, i.e.,0.03877 USD/kWh/16/ for 20 years.

The project activity involves installation of 60 MW (AC) Solar Power Plant by Prime Road Alternative (Cambodia) Co., Ltd. The project involves deploying Solar Photovoltaic panels to harness solar radiation, converting it into DC power, and subsequently utilizing inverters to transform the DC power into AC power. The objective of the project is to produce clean electricity using solar energy and channel this generated power into the National Grid of Cambodia via the electricity transmission line through power purchase agreement, there by displacing electricity generated by fossil fuel-based power plants from the National Grid of Cambodia. The project is expected to contribute 3 SDGs which are SDG 7, 8 and 13.

The project activity is commissioned on 11/11/2022. As verified from the document 'Energization and Commissioning Tests Of 60 MW Solar Photovoltaic Power Plant in Kampong Chhnang Province, Cambodia'/20/, the project was successfully energized and synchronized on 11/11/2022, which is considered as the start date of the project activity. The entire project capacity is linked to Cambodia's National Grid via established distribution and transmission lines.

The operational lifetime of project activity is 20 years, as guaranteed by the suppliers of PV panels of the project activity and same has been verified from the technical data sheet/22/, TDD/49/ and cross checked with PPA/16/ provided by the project Proponent and found acceptable.

The project activity is located in Tuek Phos district of Kampong Chhnang province, Cambodia. The location details have been verified during the onsite visit and geo coordinates verified through google earth/Maps and found to be correct.

The project is expected to achieve an annual average emission reduction of 81,868 tCO_2e over the first crediting period of 7 years and a total emission reduction of 573,078 tCO_2e during the first seven years renewable crediting period/2/.

During the current verification period, i.e., 11/11/2022 to 31/12/2023 (1st verification of 1st Crediting period), the project has achieved an emission reduction of 91,839 tCO_2e/3/.

The purpose and scope of validation and verification:
Purpose: The purpose of a validation and verification is to have a thorough and independent assessment of the project activity against the applicable VCS requirements, in particular, the project's baseline, monitoring plan and compliance with the relevant VCS and host party criteria. These are validated/verified in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation/Verification is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of emission reductions. 4KES's objective is to perform a thorough, independent assessment of the validation and verification of project activity.

Scope: Validation and Verification scope is defined as an independent and objective review of the VCS Joint Project Description and Monitoring Report (JPDMR).
The JPDMR is reviewed against the relevant criteria and guidance documents provided by VCS which include the following: VCS Program Guide (v4.4, dated 29/08/2023)/7/, VCS Standard (v4.7, dated 21/03/2024)/6/, VCS Program Definitions (v4.5, dated 16/04/2024)/4/, Registration & Issuance Process (v4.5, dated 16/04/2024)/5/, VCS Validation and Verification Manual (v3.2, dated 19/10/2016) /8/ applicable at the time in order to confirm that the project meets the applicability conditions of the selected baseline and monitoring CDM methodology ACM0002, "Grid-connected electricity generation from renewable sources", Version 21.0/9/, also assess the claims and assumptions made in the JPDMR without limitation on the information provided by the project proponents.

The method and criteria used for validation and verification:
The validation and verification consist of the following four phases:

I. A desk review of the project description and monitoring report documents
   - A review of data and information;
   - Cross checks between information provided in JPDMR and information from sources with all necessary means without limitations to the information provided by the project proponent;

II. Site Visit interviews with project stakeholders

- Interviews with relevant stakeholders in host country with personnel having knowledge with the project development via telephone, email or direct on-site visits;
- Cross checking between information provided by interviewed personnel with all necessary means without limitations to the information provided by the project proponent;

III. Reference to available information relating to projects or technologies similar to projects under validation and review based on the approved methodology being applied for the appropriateness of formulae and accuracy of calculations.

IV. The resolution of outstanding issues and the issuance of the final validation and verification report and opinion.

The number of findings raised during validation and verification:

During the course of combined validation and verification, a total of 35 findings were raised, which includes:

20 Corrective Action Requests (CARs);

15 Clarification Requests (CLs);

0 Forward Action requests (FARs);

All the raised findings have been successfully closed.

Any uncertainties associated with the validation and verification:

There are no uncertainties associated with the validation and verification of the project activity. The validation was conducted to provide a reasonableness of assumptions. The verification has been done with a reasonable level of assurance.

Summary of joint validation and verification conclusion:

'4K Earth Science Private Limited' concludes the joint validation and verification (11/11/2022 to 31/12/2023) of the first crediting period with a positive opinion that the VCS Project "Kampong Chhnang 60 MW Solar Power Project" as described in the JPDMR (version 03.0, dated 20/02/2025) /1/, meets all applicable VCS requirements, including those specified in the VCS Standard (v4.7) /6/, relevant methodology, tools and guidelines.

The selected baseline and monitoring methodology ACM0002, "Grid-connected electricity generation from renewable sources", Version 21.0/9/ is applicable to the project and is correctly applied. '4K Earth Science Private Limited' therefore requests the registration and issuance of the project activity under VCS.

CONTENTS

1 INTRODUCTION

1.1 Objective

‘Kosher Climate India Private Limited’ has commissioned ‘4K Earth Science Private Limited’ to perform the Joint validation and verification of 1st monitoring period of the project activity: “Kampong Chhnang 60 MW Solar Power Project”, with regard to the relevant VCS requirements of VCS Standard version 4.7/6/; to attain real, measurable, additional and permanent emission reductions. The monitoring period covering under this verification is from 11/11/2022 to 31/12/2023 (both dates included). The VCS projects must undergo an independent third-party validation, verification and certification of emission reductions as the basis for registration of the project activity and issuance of Voluntary Emission Reductions (VERs) from the project activity.

The purpose of the joint validation and verification is to ensure a thorough, independent assessment of the joint project description and monitoring report (JPDMR), in particular the project's description eligibility, ownership, crediting period, baseline assessment, monitoring plan by review of objective evidences.

- The requirements of VCS Program Guide Version 4.4 /7/
- The requirements of VCS Standard Version 4.7 /6/
- The requirements of applied CDM methodology ACM0002 and relevant Tools /9/ to /13/
- To assess the project’s compliance with other relevant rules, including the project country legislation
- Other relevant rules are validated and verified in order to confirm that the project description and monitoring report as documented is sound and reasonable and meet the stated requirements and identified criteria.
- The project activity has been implemented and operated as per the project description (PD) and that all physical features (technology, project equipment, and monitoring equipment) of the project are in place;
- The joint project description and monitoring report/1/ and other supporting documents/2/3/ are complete;
- The data is recorded and stored as per the monitoring methodology and approved monitoring plan.
- To establish that the data reported are accurate, complete, consistent, transparent and free of material error or omission by checking the monitoring records and the emissions reduction calculation.

The validation and verification are the requirement of all VCS projects and is seen as necessary to provide assurance about the quality of the project and its intended generation of emission reductions over the project's crediting period without any double counting. This report contains the findings and resolutions from the joint validation and verification of the project activity.

1.2 Scope and Criteria

The validation scope is given as an independent and objective review of the project design and monitoring plan, the project's baseline study and monitoring plan (ACM0002, "Grid-connected electricity generation from renewable sources", Version 21.0)/9/ which are included in the VCS PD and other relevant supporting documents are in compliance with the requirements of VCS Standard, version 4.7/6/. 4K Earth Science Private Limited has employed a risk-based approach in the validation, focusing on the identification of significant risks for project implementation and the generation of emission reductions.

The scope of the verification is the independent and objective review and ex-post determination of the monitored reductions in GHG emissions. The verification of this project is based on the MR and supporting documents submitted by the project proponent to the assessment team.

The documents were reviewed against the following guidance and protocols:

- VCS Program Guide (v4.4)/7/
- VCS Standard (v4.7)/6/
- VCS Program Definitions (v4.5)/4/
- VCS Registration & Issuance Process (v4.5)/5/
- CDM approved methodology ACM0002 (version 21.0)/9/

The scope of work covered in the joint validation and verification is described below:

- To validate whether the project activity meets the requirements of VCS Standard/6/ and VCS program guide/7/ including additionality, proof of title and compliance with local laws.
- To evaluate whether the baseline and monitoring plan are in conformance with the applied methodology from the VCS approved GHG program.
- To confirm that the information presented are completed, consistent, transparent and free of omission or material error.
- Background investigation and follow up interviews.
- Issuance of draft validation and verification report with CARs, CLs & FARs, if any.
- Final validation and verification reporting
    - The joint validation and verification is based on the information made available to 4K Earth Science Private Limited and on the contract conditions.

- The information in the VCS PDMR is reviewed against the criteria of VCS Standard, Ver 4.7/6/, VCS program guide, Ver 4.4 /7/ and the approved baseline and monitoring CDM methodology ACM0002, Version 21.0 /9/.
- 4KES has performed validation and verification based on a risk-based approach focusing mainly on the significant risks to meet the qualification criteria and the ability to generate Verified Carbon Units (VCUs).
- The validation and verification is not meant to provide any consulting to the project proponent. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project description. The work carried out by 4K Earth Science Private Limited is free from any conflict of interest.

1.3 Reasonableness of Assumptions and Level of Assurance

The assessment was conducted to provide a reasonableness of assumptions of conformance against the defined audit criteria and materiality thresholds within the audit scope. The assessment team verified the assumptions, limitations and methods for all the parameters and confirms that the reported ex-ante emission reductions are free from any type of material errors, omissions and misinterpretations.

The level of assurance of the verification report falls under reasonable assurance engagements as selected by the Client. The assessment team verified the monitoring data for all the parameters of the monitoring plan and confirms that the reported emission reductions are free from any type of material errors.

Based on the audit findings, a positive evaluation statement reasonably assures that the project GHG assertions are materially correct and is a fair representation of the GHG data and information. Before being sent to the client, the joint validation and verification report underwent an independent internal technical review to ensure that all validation and verification tasks had been carried out in accordance with the relevant 4KES requirements. Technical reviewer qualified under the 4KES's qualification scheme for VCS validation and verification conducted the technical review.

In accordance with Section 4.1.2 of the VCS Standard v4.7/6/, the joint validation and verification report is reasonableness of assumptions and level of assurance of this is reasonable.

1.4 Summary Description of the Project

The project activity involves installation of 60 MW (AC) Solar Power Plant by Prime Road Alternative (Cambodia) Co., Ltd. The project involves deploying Solar Photovoltaic panels to harness solar radiation, converting it into DC power, and subsequently utilizing inverters to transform the DC power into AC power. The objective of the project is to produce clean electricity using solar energy and channel this generated power into the National Grid of Cambodia via the electricity transmission line through power purchase agreement, there by displacing electricity generated by fossil fuel-based power plants from the National Grid of Cambodia. The project is expected to contribute 3 SDGs which are SDG 7, 8 and 13. The project activity is located in Tuek Phos district of Kampong Chhnang province, Cambodia. The location details have been verified during the onsite visit and geo coordinates verified through google earth/Maps and found to be correct

The project activity is a bid-based project awarded to Prime Road Alternative (Cambodia) Company Limited in a reverse auction tender process to supply electricity to EDC at a tariff of $0.03877 per kWh for 20 years. The same was confirmed by assessment team by checking Letter of Award/35/ and PPA/16/.

The project activity is commissioned on 11/11/2022. As verified from the document 'Energization and Commissioning Tests Of 60 MW Solar Photovoltaic Power Plant in Kampong Chhnang Province, Cambodia'/20/, the project was successfully energized and synchronized on 11/11/2022, which is considered as the start date of the project activity. The entire project capacity is linked to Cambodia's National Grid via established distribution and transmission lines.

The Project comprises 300 units of 200kVA (1500V) Huawei String inverters combined with 10 sets of box-transformers. The PV Modules are installed on single horizontal axis tracking system, consisting of 2 types of tracking row configuration. One type has 3 strings of PV modules and the another with 2 strings of PV Modules, all installed in one row portrait orientation. The operational lifetime of major project equipment is estimated to be 20 years.

There are total of 10 sub-systems, 9 of which have 6.5MVA box-transformers and the other one has a 3.25MVA box-transformer, each one connected to 4 circuits of 22kV switchgear side of the 115kV Project substation. The four (4) output of 22 kV ring main system (each comprising 20 MW output) will be connected to the Interconnection point at EDC's substation via underground cable (from the nearest box transformer to the Interconnection point). The Interconnection Point is at four (4) of 22 kV incoming feeders of EDC substation. The revenue meters are also installed at the Interconnection Point. These meters are of Landis+Gyr E650 make and have an accuracy class of 0.5s.

Incoming feeders, main and back-up meter details are as below:

Incoming *Feeder #1, main meter No. 55548513, back-up meter No. 55548515*

Incoming Feeder #2, main meter No. 55548514, back-up meter No. 55580597

Incoming Feeder #6, main meter No. 55548500, back-up meter No. 55580600

Incoming Feeder #7, main meter No. 55548498, back-up meter No. 55548501

Technical Specifications of the project activity:

Nominal DC Power of the facility	77.00616MWp

Nominal power of PV module	540 Wp
Type of PV module (Photovoltaic Panel)	540W
Module Technology	Mono-PERC Bifacial PV module
No. of modules (Total) Mounting substructure	142,604 pcs Horizontal Single Axis Tracking system: East-West trackers
Inverter	300 sets of Huawei SUN2000 - 215KTL
Smart Transformer Station	9 sets of 6500 kVA 1 sets of 3250 kVA
Ring main unit (RMU)	10 sets of 22KV RMU (available in Smart Transformer station)

Technology	Type	Manufacturer	Model
Solar PV Module	Monocrystalline bifacial	Suntech	STP540S-C72/Pmh+ (540Wp)
Inverter	String Inverter	Huawei	SUN2000-215TL-HO
Transformer	Box Transformer	Huawei	3250kVA
	Box Transformer	Huawei	6500kVA
Monitoring Equipment			
- Pyranometer	Secondary standard	Kipp & Zonen	CMP10
- PV module temperature sensor	RTD sensor	OMEGA	SA2-RTD
- Temperature & Pressure sensor & Relative humidity	Weather transmitter	Vaisala	WXT536
Mounting Structure	Horizontal single axis tracker	Clenergy	EzTracker D1P

The project is expected to achieve an annual average emission reduction of 81,868 tCO2e over the first crediting period of 7 years and a total emission reduction of 573,078 tCO2e during the first seven years renewable crediting period/2/.

During the current verification period, i.e., 11/11/2022 to 31/12/2023 (1st verification of 1st Crediting period), the project has achieved an emission reduction of 91,839 tCO2e/3/.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The combined validation and verification were conducted using 4KES's procedures in line with the requirements of VCS Standard, version 4.7/6/ and the requirements specified in the CDM M&P, the latest version of the CDM Validation and Verification Standard, version 3.0/14/, VCS validation and verification manual version 3.2/8/ and relevant decisions of the COP/MOP and the CDM EB and applying standard auditing techniques. The GHG emission reductions are on the basis of approved methodology ACM0002, "Grid-connected electricity generation from renewable sources", Version 21.0/9/.

The validation and verification consisted of the following phases:
• The desk review of documents and evidences submitted by the project proponent in context of the reference VCS rules and guidelines
• Undertaking on-site audit, interview or interactions with the representative of the project proponent
• Reporting audit findings with respect to clarifications and non-conformities and the closure of the findings, as appropriate and
• Preparing a draft validation and verification report
• Resolution of outstanding issues and the issuance of the final validation and verification report and opinion

The prepared joint validation and verification report and other supporting documents then undergo an internal quality control before being submitted to the VCS board for registration and issuance of credits as per VCS standard v4.7/6/.

Validation and Verification Contract	26/02/2024
On-site Assessment	10/05/2024
Findings raised	15/05/2024
Draft Verification Report	02/11/2024
Final Verification Report	22/02/2025

2.2 Document Review

The assessment team has conducted the validation and verification using the VCS Standard and the applied methodology ACM0002, Version 21.0/9/ as the reference criteria. The assessment team had done the completeness check of revised VCS PD MR/01/ submitted by the PP as per the VCS standard Version 4.7 /6/ requirements. Furthermore, a desk review was also carried out to assess the following:

- Information of project details in compliance with VCS PDMR/01/ template Ver 4.4
- Appropriateness of methodology ACM0002, Version 21.0/9/ applied to the project activity
- Compliance with relevant laws and regulations
- Correctness of application of baseline and monitoring methodology
- Demonstration of additionality
- Monitoring plan described in the revised VCS PD
- Compliance of monitoring plan in the MR with the revised PD.

The VCS JPDMR version 01 dated 20/02/2024 /01/ was initially reviewed and the PP was requested to submit the revised documents along with the supporting information and documents. The revised documents and additional supporting documents were further assessed by the assessment team during desk review. During the validation and verification process, the revised VCS JPDMR, version 3.0, dated 20/02/2025/01/ and the supporting documents were assessed to confirm the actions taken by the PP to the CARs and CLs issued.

The assessment team has reviewed the final version of the VCS JPDMR, version 3.0, dated 20/02/2025/01/ to confirm that all changes agreed have been incorporated adequately.

The documents that were considered during the joint validation and verification process are given in Appendix 1 of this report.

2.3 Interviews

The VVB has conducted the on-site visit for the joint validation and verification of the project activity. The on-site visit took place on 10/05/2024. The key personnel interviewed, and main topics of the interviews are summarized in the table below:

Sl. No	Name	Role/Organization	Date	Subject	Audit Team Members
1	Richat Svang	Prime Road Alternative (Cambodia) Co., LTD	10/05/2024	• Project Design and operation	Ms. Swati S Acharya (Team Leader,
2	Tian MingFei	Uper Energy Singapore PTE. LTD.			

3	Meas Charmvong	Uper Energy Singapore PTE. LTD.		• Status of implementation • Ownership • Baseline Scenario and its assumptions • Additionality • Start Date & Crediting period of the project activity • Local Stakeholder consultation • Monitoring plan and monitoring arrangements • Emission reduction calculations • SDG Contributions • Safeguards related to project activity • Grievance mechanism	Technical Expert) Mr. Sriram S (Trainee)
4	Ty Sopha	China CACS Engineering Corporation			
5	Kong Sopheak	Prime Road Alternative (Cambodia) Co., LTD			
6	Vidya Kunjappa	Kosher Climate India Private Limited			
7	Mr. Vong Sam	Chief of Royeas Village (Local)		• Related to Baseline Survey • Knowledge about Project activity • Grievances about	
8	Mr. Say Som	Prey Chrouv Village Chief (Local)			
9	Ms. Kheng Srey	2nd Vice chief of Kbal Toek commune (Local)			

				Project activity	

The solar plant is situated in Prey Chrov village, Kbal Toeuk commune, in the Tuek Phos District. The village chief of Prey Chrov, Mr. Say, the second village chief of Kbal Toeuk commune, Ms. Kheng, and Mr. Vong, the chief of Royeas village, where the project access roads pass through, are key stakeholders. Additionally, all three are members of the Grievance Redressal Committee. Hence, they were selected for interviews as representatives of the farmers and other community members of the villages that are impacted by the project.

2.4 Site Visits

The VVB has conducted the on-site visit for the joint validation and verification of the project activity. The on-site visit took place on 10/05/2024 in project site located at Tuek Phos district of Kampong Chhnang province, Cambodia.

The following was carried out during the on-site visit interviews:

- An assessment of the project design and technical specifications, project location, implementation status and operation of the project activity as per the VCS-JPDMR/01/;

- A review of information flows for generating, aggregating and reporting the monitoring parameters;

- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan in the VCS-JPDMR/01/;

- A review of calculations and assumptions made in determining the GHG data and estimated emission reductions;

- An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

The assessment team has verified sufficient appropriate audit evidence, to reduce audit risk to an acceptably low level as requisite to achieve reasonableness of assumption and reasonable level of assurance for the current validation and verification.

2.5 Resolution of Findings

The objective of this joint validation and verification is to resolve any outstanding issues which need to be clarified for 4KES's positive conclusion on the project description and monitoring report. To guarantee transparency, a validation/verification protocol has been customized for the project. The protocol shows in a transparent manner the requirements, means of validation and verification and the results from verifying the identified criteria. The validation/verification protocol consists of three tables; the different columns in these tables are described below.

A corrective action request (CAR) is raised if one of the following occurs:

- non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- Modifications to the implementation, operation and monitoring of the registered project activity has not been sufficiently documented by the project participants;
- Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impair the estimate of emission reductions;
- Issues identified in a FAR during validation/verifications to be verified during subsequent verification have not been resolved by the project participants.

A clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met. All CARs and CLs raised by the 4KES during verification shall be resolved prior to submitting a request for issuance.

FAR (Forward Action Request) is raised during this validation and verification, if the monitoring and reporting require attention and/or adjustment for the next verification period.

In summary, 15 CLs and 20 CARs were raised and resolved satisfactorily. No FAR has been raised in the verification. The list of CARs/CLs/FARs raised and the response provided, the mean of validation, reasons for their closure and references to correction in the relevant documents are provided in Appendix III of this report.

2.5.1 Forward Action Requests

No FAR is raised during the Joint Validation and Verification process.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity involves installation of 60 MW (AC) Solar Power Plant by Prime Road Alternative (Cambodia) Co., Ltd. The project involves deploying Solar Photovoltaic panels to harness solar radiation, converting it into DC power, and subsequently utilizing inverters to transform the DC power into AC power. The objective of the project is to produce clean electricity using solar energy and channel this generated power into the National Grid of Cambodia via the electricity transmission line through power purchase agreement, there by displacing electricity generated by fossil fuel-based power plants from the National Grid of Cambodia. The project is expected to contribute 3 SDGs which are SDG 7, 8 and 13.

The project employs the approved CDM large-scale methodologies ACM0002, “Grid-connected electricity generation from renewable sources”, Version 21.0/9/. The project activity is commissioned on 11/11/2022. As verified from the document ‘Energization and Commissioning Tests Of 60 MW Solar Photovoltaic Power Plant in Kampong Chhnang Province, Cambodia’/20/, the project was successfully energized and synchronized on 11/11/2022, which is considered as the start date of the project activity. The entire project capacity is linked to Cambodia’s National Grid via established distribution and transmission lines.

Additionally, the project proponent has entered into long-term Power Purchase Agreements (PPA) with Electricite du Cambodge (EDC) to provide the generated power at the agreed-upon price, i.e.,0.03877 USD/kWh/16/ for 20 years.

The project is expected to achieve an annual average emission reduction of 81,868 tCO2e over the first crediting period of 10 years and a total emission reduction of 573,078 tCO_2e during the first seven years renewable crediting period/2/.

During the current verification period, i.e., 11/11/2022 to 31/12/2023 (1st verification of 1st Crediting period), the project has achieved an emission reduction of 91,839 tCO_2e/3/.

Hence, validation team confirms that the description in the project description is accurate, complete, and provides an understanding of the nature of the project.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	The assessment team has checked the audit history mentioned under section 1.2 of the JPDMR /1.0/ and found that the JPDMR /1.0/ has been filled appropriately.
Sectoral scope	Sectoral scope of the project activity: Sectoral Scope 1: Energy (renewable/non-renewable) As the project activity involves generation of electricity through solar power (renewable energy) and supplying the generated electricity to the national grid of Cambodia, assessment team confirms that the sectoral scope of the project activity has been correctly considered, thus deemed acceptable.
AFOLU project category, if applicable	Not Applicable as the project activity is not an AFOLU project.
Project activity type	The registered project activity falls under ‘Renewable energy Project’ category which is correct, as the project activity involves generation of electricity through solar power

	(renewable energy) and supplying the generated electricity to the national grid of Cambodia.
General eligibility of the project to participate in the VCS Program	1) The Project is a 60 MW grid connected electricity generation activity using solar power in Cambodia, which is a Least developed Country (LDC) as designated by the United Nations. As per VCS Standard v4.7, section 2.1 - Table 1, grid connected solar power plants located in LDCs are within the VCS program scope, making this project eligible under VCS. 2) The start date of the project activity is 11/11/2022 which is the commissioning certificate/20/ of the project activity, i.e., the day from which the project began generating GHG emission reductions which is in-line with the requirement set out in paragraph 3.8 of VCS Standard Version 4.7/6/. 3) VCS Standard v.4.7/6/ requires the use of one of the VCS approved CDM methodology ACM0002, version 21.0/9/. The Project activity uses the tools and methodologies approved by UNFCCC CDM executive board, which is a VCS approved program. 4) The project does not benefit from any kind of Official Development Assistance. 5) In addition to the above, the project is not a grouped project activity, nor de-bundled part of a grouped project activity Hence, the project is eligible for registration under VCS program as per VCS Standard v4.7/6/.
AFOLU project eligibility, if applicable	Not Applicable as the project activity is not an AFOLU project.
Transfer project eligibility, if applicable	Not Applicable as the project activity is not a transfer project.
Project design	The project activity is in a single location and is a standalone project. This was confirmed by the Assessment team during the on-site audit.
Project ownership	The ownership of the project activity lies with Prime Road Alternative (Cambodia) Co., Ltd. The same was verified by the assessment team by reviewing the tender award certificate/35/, land lease agreement/34/, Environmental Impact Assessment (EIA) license/26/, equipment purchase contract/21/ and the power purchase agreements/16/.

Project start date	Project start date is 11/11/2022, it is the date from which the project activity starting generating emission reductions. The considered date is the commissioning date of the project activity. The same was verified by the assessment team from the commissioning certificate/20/ of the project activity. Thus, assessment team confirms that the PP has considered the start date appropriately as per section 3.8.1 of the VCS Standard 4.7/6/.
Project crediting period	As verified from the JPDMR/1/, the crediting period is from 11/11/2022 to 10/11/2029. PP has chosen 07 years Crediting period, renewable twice. As per section 3.9.1 of VCS Standard, version 4.7/6/, the crediting period is accepted to the assessment team.
Project scale	< 300,000 tCO_2e/year (project) The emission reductions from the project are less than 300,000 tonnes of CO_2e per year, hence the project scale comes under the category of "Project" (as per para 3.10.1 of the VCS standard Version 4.7/6/). There is no change in the project scale since the commissioning.
Likelihood of achieving estimated GHG emission reduction or removals	During the 1st crediting period, the project is expected to produce 141,719 MWh/yr electricity (with annual degradation of 0.45%) and a total amount of 81,868 tonnes of CO_2e GHG reductions each year. Total estimated emission reduction within the 1st crediting period is 573,078 tCO_2e. This was verified from the ER calculations provided in the ER sheet/2/.
Technologies and measures implemented by the project activity	The technologies and measures implemented by the project activity is included in section 1.12 of the JPDMR/1/. The same was verified by the assessment team by means of on-site inspection, checking the name plate details and technical specifications from the manufacturers' manual/datasheet/22/. Hence, assessment team concludes that the technologies employed have been correctly included in the JPDMR/1/.
Implementation schedule of the project activity or activities	According to the VCS Standard, Ver 4.7, the project start date is considered as the date on which project started generating emission reductions. As per the VCS JPDMR/01/, the project activity is fully commissioned and started its operation on

	11/11/2022. The same has been verified from the commissioning certificate/20/ and found to be correct.
Project location	The project activity is located in Tuek Phos district of Kampong Chhnang province, Cambodia. The geo-coordinates of the project activity are as below: Latitude: 104°23'50"E; Longitude: 11°47'31"N The location details have been verified during the onsite visit and geo coordinates verified through google earth/Maps and found to be correct and consistent with the details given in section 1.13 of JPDMR/01/.
Conditions prior to project initiation	The project activity is a greenfield solar power plant/facility, where no project was available at the site of the project proponent prior to the implementation of this project activity. Thus, in pre-project scenario, no energy generation sources existed in this location. The project proponent has implemented solar power project to generate electricity and supply the generated electricity to the national grid of Cambodia.
Project compliance with applicable laws, statutes and other regulatory frameworks	The compliance of project activity with the applicable laws, statutes and other regulatory framework has been included in the section 1.15 of the JPDMR/01/. The same was assessed by the audit team and found to be correct. Hence, assessment team concludes that the project activity complies with all the applicable laws of the host country.
Double counting and participation under other GHG programs	The project is registered under VCS only and has not been rejected by any other GHG Program. No registration or issuance is sought for this validation and verification period under any other mechanism and the declaration is given by the PP for the same/28/. The Assessment team confirms that it has checked that there is no double counting associated with project activity being participation of other GHG programs. Declaration /28/ for the same has been submitted.

	Thus, emission reductions generated by the project will be solely claimed by PP and PP has the right of use, which is acceptable.
No double claiming with emissions trading programs or binding emission limits	Net GHG emission reductions generated by this project activity will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions as the host country. Also, the project has not sought or received another form of GHG-related environmental credits including renewable energy certificates. The same is confirmed from the interview with PP and their representative and the no-double-counting declaration provided by PP/28/.
No double claiming with other forms of environmental credit	Project activity has not sought, received, or is planning to receive credit from another GHG-related environmental credit system. The same is confirmed from the interview PP and their representative and the no-double-counting declaration provided by PP/28/.
Supply chain (Scope 3) emissions double claiming	Project proponent or authorized representative is not a buyer or seller of a product whose emissions footprint is changed by the project activities. The same is confirmed from the interview with PP and their representative.
Sustainable development contributions	Assessment team has checked the supportive for the activities that result in the SD contributions from the project activity for each SDG. **SDG Target 7.2:** By 2030, increase substantially the share of renewable energy in the global energy mix. **SDG Indicator 7.2.1:** Renewable energy share in the total final energy consumption During the current monitoring period, 156,242 MWh of electricity was generated from the renewable energy source. The assessment team has verified the JMR readings and invoices and found consistent. **SDG Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value** **SDG Indicator 8.5.1: Average hourly earnings of employees, by sex, age, occupation and persons with disabilities** During the current monitoring period, the project activity provided jobs for 42 people. In total, currently, a total of 42 people are employed for the project activity. The assessment

	team has verified the same during the interviews and list of employees from the HR Records and declaration provided by PP/30/. Project activity provide local jobs. Section 1.18 of the JPDMR report has been checked and found OK. The project activity contributes positively to sustainable development which has been verified from the interviews of local people (Section 2.3 of this report) and supportive documents as mentioned above. **SDG Target 13.2: Take urgent action to combat climate change and its impacts** **SDG Indicator 13.2.2: Tonnes of greenhouse gas emissions avoided or removed** During the monitoring period, project has avoided the emission of 91,839 tCO_2e. The assessment team has checked the calculation from the JPDMR /1/ and the Actual ER calculation sheet /3/ and found correct. Sustainable development contributions from the project activity have been mentioned under section 1.18 of the VCS JPDMR/01/ and found OK. The Assessment team has interviewed few stakeholders and did not find any complaints. They were having positive opinion about the project in terms of job creation for locals and other infrastructure development in the project area. The logbooks and employment records were verified to confirm the contribution of the project during the monitoring period.
Additional information relevant to the project	No commercially sensitive information has been excluded from the public version of the monitoring report.

3.2 Project Activity Instances in Grouped Projects

This is not applicable, as the project activity is not a grouped project.

3.3 Safeguards

3.3.1 Stakeholder Engagement and Consultation

3.3.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion

Stakeholder identification	The details on identification of stakeholders is provided in section 2.1.1 of JPDMR. The following are the key relevant stakeholders identified: • Local communities, individuals, and their representatives who are anticipated to be either directly or indirectly impacted by the project or have an interest in its outcomes. • Local policymakers and representatives from local authorities who are involved in or influenced by the project's development. • Local non-governmental organizations (NGOs) and women's groups that focus on issues pertinent to the project or collaborate with communities likely to be affected. As the list includes all the people impacted directly or indirectly by the project activity is identified by PP, the same has been acceptable to the assessment team. Assessment team has confirmed the identified stakeholders from the site visit, Minutes of stakeholder consultation, presentation, photos and attendance sheet, PP representative and local people interview and confirmed that the identification of stakeholders is in line with section 3.18.1 paragraph 1 of VCS standard 4.7.
Legal or customary tenure/access rights	There are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, Indigenous People (IPs), local communities (LCs), and customary rights holders. The project is implemented on government-owned (EDC) land. However, these lands have been leased on a long-term basis by the project proponent. The assessment team has verified this by reviewing the land lease records/34/ of the project. As a result, no other entities hold legal or customary tenure or access rights to the territories and resources within the project area. Assessment team confirmed that there is no conflict as confirmed from local people interviews.
Stakeholder diversity and changes over time	The details on Stakeholder diversity and changes over time is provided in section 2.1.1 of JPDMR. The stakeholders come from various diverse background as observed from the list of stakeholders invited for the stakeholders meeting and diversity is unlikely to change over time. Assessment team confirm the same by interviewing PP, their representatives and local people.

Expected changes in well-being	The details on Expected changes in well-being is provided in section 2.1.1 of JPDMR. The project is developed to utilize clean energy for electricity generation and creates temporary and permanent jobs with good working conditions for local people. The same is confirmed during the site visit by interviewing PP, employee interviews and list of employees /30/.
Location of stakeholders	The details on location of stakeholders is provided in section 2.1.1 of JPDMR. Assessment team has confirmed the same from the PP representative and Local people interviews. Further, it was also observed that there is no change in stakeholders during this monitoring period.
Location of resources	There are no habitations/settlements within the project site. This site was previously used for cassava plantations and was later reportedly abandoned. The site assessment conducted by the project proponent for 97 ha identified the site as mainly scrubland and paddy fields. The site surrounding the project, is now a grass-field and a grazing area for local villagers. This is a kind of opportunistic grazing by villagers. There are no territories or resources in the project site which are owned by the stakeholders or have customary access to. Assessment team has confirmed the same from the PP representative and Local people interviews. Further, it was also observed that there is no change in location of resources during this monitoring period.

3.3.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	The detailed Stakeholder engagement process is provided in section 2.1.2 of JPDMR/01/. The stakeholder consultation process is designed to be robust, inclusive, and transparent, ensuring that all key stakeholders are identified and invited. Invitations were verified against a comprehensive local stakeholder list, guaranteeing broad representation. Meetings bring together a diverse group of participants, including local community members, local authorities, village heads, and workers. This inclusive approach embraces cultural diversity and integrates valuable local knowledge, fostering meaningful engagement and well-informed decision-making.

	Furthermore, the project proponent has established a Grievance Redressal Mechanism (GRM) which provides an accessible platform for receiving and facilitating resolution of grievances from affected stakeholders. The same was confirmed by assessment team by checking chapter 5 of ESIA report/27/, Stakeholder Engagement Report/32/ and through interviewing PP, their representatives and local stakeholders during the onsite inspection.
Consultation outcome	The outcome of Stakeholder engagement process is provided in section 2.1.2 of JPDMR/01/. The LSC meeting was held on 21/07/2022 during the construction period to provide updates on project progress, assess its impact on the community, and offer stakeholders an opportunity to voice their concerns, share recommendations for improvement, and provide feedback. As detailed in the stakeholder consultation report, the representative of project proponent explained technical aspects and about Social, Environmental benefits including electricity generation by clean energy and job creation. Furthermore, the local stakeholders were asked to provide feedback on the project activity, including whether the project will have a positive, negative, or no impacts. The same was confirmed by assessment team by checking chapter 5 of ESIA report/27/, Stakeholder Engagement Report/32/ and through interviewing PP, their representatives during the onsite inspection.
Ongoing communication	PP has a mechanism for on-going communication with local stakeholders which has been discussed during physical site visit through interviewing PP, their representatives and local stakeholders during the onsite inspection. The detailed process for ongoing communication with the stakeholders is provided in section 2.1.2 of JPDMR/01/. During current monitoring period there are no grievances received related to the solar power plant which was confirmed during the interview of PP representative and local stakeholders. Thus, assessment team is of the opinion that the ongoing stakeholder mechanism is adequate and appropriate.
Stakeholder input	The inputs obtained from local stakeholders are in section 2.1.2 of JPDMR/1/. The same was confirmed by assessment team by checking relevant documents and through interviewing PP, their representatives and local stakeholders during the onsite inspection.

3.3.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	The detailed process for obtaining consent is provided in section 2.1.3 of JPDMR/1/. The same was confirmed by assessment team by checking relevant documents and through interviewing PP, their representatives and local stakeholders during the onsite inspection.
Outcome of FPIC discussion	The result of the FPIC and stakeholder consultation is positive, with stakeholders expressing their support for the project. Additionally, a third-party assessment was conducted to confirm that the land acquisition adhered to the ADB SPS (2009) requirements. The verification affirmed that EDC acquired the land through commercial transactions, ensuring that the project did not encroach on land, relocate people without consent, or cause forced physical or economic displacement. The same was confirmed by assessment team by checking relevant documents and through interviewing PP, their representatives and local stakeholders during the onsite inspection.

3.3.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	The development process for grievance redress procedure is detailed in section 2.1.4 of Joint PDMR/1/. The same was confirmed by assessment team by checking ESIA report and through interviewing PP and their representatives during the onsite inspection.
Grievance redress procedure	The grievance redress procedure is detailed in section 2.1.4 of Joint PDMR/1/. The same was confirmed by assessment team by checking ESIA report and through interviewing PP and their representatives during the onsite inspection.

3.3.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion

No comments were received during the current monitoring period.	Since there were no comments received during the current monitoring period, there are no updates to project design. Also, no comments were received during the public comment period which requires any action in the current monitoring period. There were no other major comments or protests raised by the stakeholders till now. The local stakeholders were totally in support of setting up of the project in the region.	The assessment team has confirmed during the on-site visit that there are no updates or retrofit done to the project. The assessment team found, through stakeholder interviews, no ongoing or unresolved conflicts, and the project does not impact conflict outcomes.

3.3.2 Risks to Local Stakeholders and the Environment

3.3.2.1 Management Experience

The PP has hired the management team of the project activity with relevant expertise/experience. The project proponent doesn't have any partnership with other organizations to support the project or have a recruitment strategy as the management personnel were hired based on the required expertise. The same was confirmed through interviewing the PP, their representatives and site personnels during onsite audit.

3.3.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	NA, as no risks are identified
Risks to stakeholder participation	There was no risk to stakeholder participation, which is confirmed by interviewing the stakeholders during the site visit.
Working conditions	The working conditions were maintained well by implementing the policy and the same is confirmed by interviewing the plant workers during the site visit

Safety of women and girls	The safety of women and girls were maintained as per the national laws on women safety in work place the same is confirmed by Assessment team in the project site during the site visit.
Safety of minority and marginalized groups, including children	The project proponent (PP) confirms adherence to local laws to ensure the safety of minority and marginalized groups, including children, at the project site. The same is confirmed by interviewing the employees on the project site.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	The implementation and operation of the project activity i.e., the solar power plant which is renewable energy project results in minimal pollution of air, noise, and water. All the waste generated from the project activity will be monitored, handled, stored and disposed based on the Waste and Wastewater management plan/56/ by PP for the project activity. The project ensures proper handling, storage, and disposal of redundant/broken panels through designated broken PV storage room at the project site. PP's Waste and Wastewater management plan was checked and found that PP records the monthly Waste generation and waste water generation and the disposal complies with the Sub-decree No. 36 on Solid Waste Management (MOE, 1999), utilizing licensed vendors for collection, transportation, and recycling. The same was confirmed by assessment team during the site visit.

3.3.3 Respect for Human Rights and Equity

3.3.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	During the site visit, it was confirmed that there were no instances of discrimination or sexual harassment within the project area. The project developer has implemented a non-harassment policy that prohibits any form of discrimination or harassment based on gender, age, religion, or other protected characteristics. Employees are informed of these policies through training programs, and clear reporting mechanisms are established to address any concerns or incidents promptly. It was

	also verified that local laws on these matters are strictly followed. This confirmation was obtained through interviews with stakeholders and company representatives.
Sexual harassment	During the site visit, it was confirmed that there were no instances of discrimination or sexual harassment within the project area. The project developer has implemented a non-harassment policy that prohibits any form of discrimination or harassment based on gender, age, religion, or other protected characteristics. Employees are informed of these policies through training programs, and clear reporting mechanisms are established to address any concerns or incidents promptly. It was also verified that local laws on these matters are strictly followed. This confirmation was obtained through interviews with stakeholders and company representatives.
Equal pay for equal work	PP is dedicated to fostering an inclusive and diverse workforce where everyone has an equal opportunity to succeed. PP is committed to offering equal opportunities for all individuals, regardless of gender, by promoting equitable pay and fair treatment in the workplace. This commitment was confirmed during the on-site visit to the project activity, where the assessment team interviewed employees.
Gender equity in labor and work	During the site visit, the documents submitted for SDG 8 were reviewed and confirmed to show no evidence of gender discrimination in job provision or wage allocation. Additionally, the project developer has implemented a Non-Discrimination and Equal Opportunity Policy, which promotes equal opportunities, non-discrimination, and fair treatment of workers.
Forced labor	During the site visit, PP provided assurance and detailed explanation regarding the background verification process for each employee. This process ensures that there are no instances of forced labor within the workforce.
Child labor	During the site visit, PP provided assurance and detailed explanation regarding the background verification process for each employee. This process ensures that there are no instances of child labor within the workforce.
Human trafficking,	During the site visit, PP provided assurance and detailed explanation regarding the background verification process for

	each employee. This process ensures that there are no instances of human trafficking within the workforce.

3.3.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk has been identified	The project aligns with applicable international human rights law, including the United Nations Declaration on the Rights of Indigenous Peoples and the International Labour Organization (ILO) Convention 169 on Indigenous and Tribal Peoples, demonstrating its commitment to promoting the rights and welfare of IPs, LCs, and customary rights holders. The project avoids any activities that could violate the economic, social, and cultural well-being of local communities. Consequently, the project developer and the project itself commit to respecting internationally proclaimed human rights. They will not be complicit in any form of violence or human rights abuses, as defined in the Universal Declaration of Human Rights. This commitment reflects the project's dedication to upholding ethical standards and safeguarding the well-being of the communities involved. This was confirmed through interview with stakeholders and company representatives during the site visit, and by verifying PP's policy and grievance record book and located in the project area.

3.3.3.3 Indigenous Peoples and Cultural Heritage

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk has been identified	Not applicable as this project will neither directly nor indirectly impact the dignity, human rights, livelihood systems, or culture of Indigenous Peoples (IPs). It will also not affect the territories or natural and cultural resources that IPs own, use, occupy, or claim as their ancestral domain. Furthermore, there are no risks to ecosystems due to the project activities, and measures will be implemented to ensure that there are no negative impacts on ecosystems. By prioritizing cultural heritage preservation, the project upholds its commitment to responsible and sustainable development practices.

3.3.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk has been identified	This project is a component of Electricite du Cambodge's (EDC) National Solar Park (NSP) initiative. The project proponent has secured a 97-hectare lease for a 60 MW photovoltaic (PV) plant from EDC through the reverse bidding process/35/. EDC acquired the land through commercial negotiations, following a willing buyer-willing seller approach. As a result, the project site does not involve any territories or resources to which the stakeholders have customary rights or access.

3.3.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	Not Applicable
Summary of the benefit sharing plan	There is no benefit sharing in the project, all the environmental rights are exclusive to the PP. The same has been verified by checking the PDMR by assessment team. Therefore, this section is not applicable.
Approval and dissemination of benefit sharing plan	Not Applicable
Benefit sharing during the monitoring period	Not Applicable

3.3.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion

Impacts on biodiversity and ecosystems	During the monitoring period, no adverse effects on biodiversity or ecosystems were observed in the Project. ESIA has confirmed that the projected area is the not located in sensitive ecological zones, biodiversity conservation areas, and there are no rare and valuable plant and animal species. No impact on biodiversity is likely to be caused, the same is verified and confirmed by audit team during the site visit.
Soil degradation and soil erosion	During the monitoring period, no instances of soil degradation or erosion were identified in the project area. The project might lead to soil erosion during the construction phase. However, mitigation measures will be implemented, such as recovering topsoil stripped during clearing and construction. Land clearing will be confined to the specific project site needed for development, minimizing the impact. Activities involving chemicals or hydrocarbons, including the storage of broken or redundant solar panels, will be limited to designated areas with proper bunding, a concrete or compacted earth base, and complete containment by drainage controls. Furthermore, soil quality monitoring and testing was carried out for six months during the construction period.
Water consumption and stress	Throughout the monitoring period, no concerns regarding water consumption or stress were identified within the project. The project will not contaminate or impede the natural flow of any surface or sub-surface water sources. Additionally, the project incorporates a water harvesting procedure to collect stormwater and drained into water retention ponds. Furthermore, ground and surface water quality tests were conducted both during the construction and operation phase.

3.3.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	Project activity is not affecting any rare habitat, the same is conformed during the site visit
Areas needed for habitat connectivity	The project is not located in or adjacent to habitats for rare, threatened, or endangered species, it is confirmed in the project location during the site visit.

3.3.4.2 Introduction of Species

No aquatic or terrestrial species are introduced as part of the project activity.

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
NA	Since the project activity is a renewable energy project, this section is not applicable

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
NA	Since the project activity is a renewable energy project, this section is not applicable

	Evidence gathering activities, evidence checked, and assessment conclusion
Invasive species	Since the project activity is a renewable energy project, this section is not applicable

3.3.4.3 Ecosystem conversion

The project activity has no impact on ecosystems, and no ecosystem conversion is possible as part of the project activity. Hence, not applicable.

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	Since the project activity is a renewable energy project, this section is not applicable

3.4 Application of Methodology

3.4.1 Title and Reference

The assessment team confirms that selected baseline and monitoring methodology is a VCS approved CDM methodology, and PP has correctly used it. The applied methodology and tools, and the specific versions of them applied in the project by the PP are valid at the time of submission of this project activity for registration.

The project activity applies the latest version of the methodology: ACM0002: Grid-connected electricity generation from renewable sources, version 21.0. /9/

The methodology and tools referenced in the project activity are given below:

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	21.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0.0
Tool	TOOL27	Investment analysis	13.0
Tool	TOOL24	Common Practice	03.1

3.4.2 Applicability

The project activity applies the large-scale Approved Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources, version 21.0 /9/. All the applied methodology and tools used for the project activity are valid at the time of submission of documents for registration. The applicability criteria for the methodology are assessed by the assessment team by means of document review and interviews. The audit team confirms that the project activity meets the criteria of the applied methodology. The applicability conditions of the methodology and the methodological tools applied are justified in the table given below:

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002	This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant;	The project involves installation of 60 MWAC Solar Power Plant Prime Road Alternative (Cambodia) Co., Ltd/20/, at a site where there was no renewable power plant operating prior to implementing the project activity (Greenfield plant).

	(b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The electricity generated from project activity is exported to the National Grid in Cambodia through power purchase agreement/16/ with EDC. Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/ as it satisfies criterion (a).
ACM0002	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic1 or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).	This is not applicable as the project activity is the installation of solar PV panels to generate electricity and does not involve the integration of a Battery Energy Storage System (BESS). Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/.

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Table 2. Combinations of renewable energy technologies and mode of BESS applicable for integration

<table>
<tr><th>Renewable Energy Technology
Mode of installation of BESS</th><th>Solar photovoltaic or wind</th><th>Other renewable technologies</th></tr>
<tr><td>BESS + (a) Greenfield plant(s)</td><td>Eligible</td><td>Eligible</td></tr>
<tr><td>BESS+ capacity addition to existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
<tr><td>BESS with no other changes to the existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
<tr><td>BESS + retrofit to existing plant(s)</td><td>Eligible</td><td>Not eligible</td></tr>
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<tr>
<td>ACM0002</td>
<td>The methodology is applicable under the following conditions:

(a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power</td>
<td>This is not applicable as the project activity is the installation of solar PV panels to generate electricity without BESS integration.

Hence the project is in compliance with latest</td>
</tr>
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	plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy	version of the methodology ACM0002 Version 21.0/9/.

	power plant(s). Only during exigencies2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.	
ACM0002	In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b). The project activity is implemented in existing single or multiple reservoirs,	This is not applicable as the project activity is the installation of solar PV panels to generate electricity. Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/.

	where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m2; or (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m2; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m2, all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m2; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m2 shall be: a. Lower than or equal to 15 MW; and	

	b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
ACM0002	In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five	This is not applicable as the project activity is the installation of solar PV panels to generate electricity. Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/.

	years prior to the implementation of the CDM project activity.	
ACM0002	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity is the installation of solar PV panels to generate electricity and it not does not involve switching from fossil fuels to renewable energy sources at the site of the project activity and installation of biomass fired power plant. Hence this applicability criterion is applicable or relevant for the project activity. Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/.
ACM0002	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	This the new installation of Solar Power Plant and not a retrofit, rehabilitations replacement or capacity additions which was verified and confirmed through onsite verification and interviewed with project proponent and their representatives. Hence it is not applicable to the project activity. Hence the project is in compliance with latest version of the methodology ACM0002 Version 21.0/9/.

Methodological Tools	Applicability condition	Assessment and conclusion

TOOL07	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	This project involves electricity generation from the solar PV modules that generate electricity and subsequently export to grid. In the absence of the project activity, the equivalent amount of power would have been drawn from the National Grid of Cambodia which is dominated by fossil fuel power plants. The baseline emissions are calculated from electricity supplied to the grid by the project activity multiplied with emission factor of the National grid. The emission factor can be calculated using OM, BM and CM using this tool. Thus, the applicability criterion is met.
TOOL07	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power	The project has selected the Default Emissions Factor (DEF) issued by the IFI Technical Working Group (IFI TWG)/43/ as the emission factor for the grid of the project. After reviewing all publicly available sources for the Emissions Factor value, it has been confirmed that the most recent version, which can be obtained prior to validation (in accordance with the requirements of CDM Tool 07: "Tool to Calculate the Emission Factor for an Electricity System")/10/, is the IFI Dataset of Default Grid Factors v.3.2, published in April 2022/43/. Therefore,

	plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	the selected data is in-line with Tool 07/10/, and hence the criteria have been met.
TOOL07	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country	The project is located on the host country Cambodia, which is not Annex I country, hence the criterion is not applicable.
TOOL07	Under this tool, the value applied to the CO2 emission factor of biofuels is zero	There are no biofuel power plants in the Host country, hence the condition is not applicable.
TOOL01	The use of the "Tool for the demonstration and assessment of additionality" is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The methodology is approved in CDM and the tool is included by the same approved methodology viz., ACM0002: "Grid-connected electricity generation from renewable sources", Version 21.0/9/. Thus, the application of this tool was found to be acceptable, and the applicability criterion is met. The project Proponent does not propose any new methodologies to demonstrate additionality.
TOOL01	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory	The methodology is approved in CDM and the tool is included by the same approved methodology viz., ACM0002 Version 21.0/9/. Thus, the application of this tool was found to be

		acceptable, and the applicability criterion is met.
TOOL27	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	Project activity applies “Tool for the demonstration and assessment of additionality”/11/. Hence this tool is applicable.
TOOL27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied approved baseline and monitoring methodology does not contain requirements for the investment analysis that are different from those described in this methodological tool. Hence this clause not applicable.
TOOL24	This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for	Project activity applies “Tool for the demonstration and assessment of additionality”/11/. Hence this tool is applicable.

	the demonstration of additionality.	
TOOL24	In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	This clause is not applicable. The applied approved baseline and monitoring methodology does not define any different approaches for the conduction of the common practice test from those described in this methodological tool.

3.4.3 Project Boundary

As per the applied methodology ACM0002 “Grid-connected electricity generation from renewable sources”, Version 21.0/9/, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to. The components of the project boundary mentioned in the JPDMR/01/ were found to be in compliance with para 22 of the applied methodology.

The assessment team conducted desk review of the implemented project to confirm the appropriateness of the project boundary identified. The assessment team confirmed that all GHG sources required by the methodology have been included within the project boundary.

It was assessed that no emission sources related to project activity will cause any deviation from the applicability of the methodology or accuracy of the emission reductions.

The project boundary is clearly depicted with the help of a pictorial depiction in section 3.3 of the Joint PDMR/1/ and duly verified by the assessment team via commissioning certificate/20/ of the project activity & power purchase agreement/16/ between project proponent and EDC which is found to be acceptable and appropriate.

Source		Gas	Included?	Assessment and conclusion
Baseline	CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO_2	Yes	Main Emission Source. The baseline scenario involves generating an equivalent amount of electricity from grid-connected fossil fuel-dominated power plants, which will lead to CO_2 emissions.
		CH_4	No	Minor Emission source.

Source		Gas	Included?	Assessment and conclusion
				Excluded for simplification. This is conservative.
		N_2O	No	Minor Emission source. Excluded for simplification. This is conservative.
		Other	No	N/A
Project	Greenfield Solar Power Project activity	CO_2	No	The project activity involves generation of electricity through solar power (renewable energy) plant. No emissions are emitted from the project as there is no emission source present at the project site. As per the applied methodology, ACM0002, project emissions are considered zero for the renewable energy power generation projects. Hence acceptable.
		CH_4	No	
		N_2O	No	
		Other	No	

In summary, the project boundary was correctly identified in accordance with the methodology ACM0002, version 21.0/9/. All greenhouse gas emissions occurring within the project activity boundary as a result of the implementation of the proposed VCS project activity have been appropriately addressed in the VCS JPDMR/01/.

3.4.4 Baseline Scenario

As per paragraph 24 of the applied methodology ACM0002 version 21.0 /9/ "If the project activity is the installation of a Greenfield power plant with or without a BESS as described under paragraph 4(a) or paragraph 5(a), the baseline scenario is electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07/10/".

The project activity involved setting up Solar power plant to harness the power of solar energy to produce electricity and supply to the grid. In the absence of the project activity, the electricity would have been generated from operation of grid-connected power plants and by the addition of new generation sources into the grid. Hence, the baseline for the project activity is the equivalent amount of power from the national grid of Cambodia which is found to be appropriate and acceptable to assessment team.

During the implementation of the project activity and establishing the baseline scenario, Project Proponent has taken into account the relevant national and/or sectoral policies, regulations and circumstances. The project activity, which involves the implementation of solar energy-based power projects for electricity generation, is not compulsory or mandated by any legal requirement in Cambodia. Instead, it is a voluntary undertaking initiated by the involved parties.

As per paragraph 24 of the applied methodology, baseline emissions include only CO2 emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are the product of Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in project year expressed in MWh multiplied by the grid emission factor in tCO_2/MWh.

Determination of Grid Emission Factor ($EF_{grid, CM, y}$):

Parameter	Value (tCO_2/MWh)	Source	Assessment and Conclusion
Combined margin CO_2 emission factor for the project electricity system in year y: $EF_{grid, CM, y}$	0.5878	Combined margin Emission factor is sourced from the IFI Technical Working Group (IFI TWG) dataset containing Default Emission Factors for electricity grid, and is sourced from the most recent version of the Harmonized IFI Default Grid Factors 2021 (Version 3.2)	The project has selected the Default Emissions Factor (DEF) issued by the IFI Technical Working Group (IFI TWG)/43/ as the emission factor for the grid of the project. After reviewing all publicly available sources for the Emissions Factor value, it has been confirmed that the most recent version, which can be obtained prior to validation (in accordance with the requirements of CDM Tool 07: "Tool to Calculate the Emission Factor for an Electricity System"), is the IFI Dataset of Default Grid Factors v.3.2, published in April 2022/43/. Based on the aforementioned source, it has been verified that the latest $EF_{grid, CM, y}$ for Intermittent energy (e.g., solar, wind, tidal) is 0.8743 tCO_2e/MWh and Firm energy (e.g., hydro, geothermal) is 0.5878 tCO_2e/MWh in Cambodia for 2021. The Project Participant (PP) has selected an emission factor of 0.5878 tCO_2e/MWh to calculate the emission reductions for the project as a conservative approach, which is acceptable to the assessment team.

Hence the validation team confirms that the proposed project activity meets the tool and methodological requirement. Therefore, the identified baseline scenario is applicable to the proposed project activity

All the assumption and data used by the project participants are listed in the VCS PDMR and/or supporting documents. All documentation relevant for establishing the baseline scenario are correctly quoted and interpreted in the VCS PDMR. Assumptions and data used in the identification of the baseline scenario are justified, appropriately supported by evidence and can be deemed reasonable. Relevant national and/or sectoral policies and circumstances are considered and listed in the VCS PDMR.

The approved baseline methodology has been correctly applied to identify a realistic and credible baseline scenario, and the identified baseline scenario most reasonably represents what would occur in the absence of the proposed VCS project activity.

3.4.5 Additionality

The PP has adopted the stepwise approach for demonstrating and assessing the additionality of the project activity as follows

Step 0: Demonstration whether the project activity is the first-of-its-kind.

This step is optional and not used for this project as this is not a first of its kind project activity.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity. These alternatives are as below:

Alternative (a) The proposed project activity undertaken without being registered as a VCS project activity.

Alternative (c) Continuation of the current situation (no project activity or other alternatives undertaken).

Outcome of Step 1a: Alternative (a) & (c) are realistic and credible alternative scenario(s) to the project activity.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative(s) shall be in compliance with all applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution.

Under Cambodia's Electricity Law, there are no legal restrictions that prevent the implementation of renewable energy projects, ensuring that alternative (a) is fully in line with the country's laws and regulations.

Alternative (b), which reflects the continuation of the baseline scenario where electricity supplied to the EDC grid by the project would have otherwise been generated by grid-connected power plants or additional generation sources, also complies with Cambodian regulations. The

Electricity Law of the Kingdom of Cambodia does not prohibit the grid from delivering the same amount of electricity annually.

Step 2: Investment Analysis

Under step 2, it is demonstrated that project activity is not economically or financially feasible, without the revenue from the sale of approved carbon credits. Further to conduct the investment analysis, Methodological tool: Investment analysis, version 13.0/12/, has been referred which is appropriate and acceptable to verification team also in line with the paragraph 97 of VVS Version 3.0.

Sub-step 2a: Determine appropriate analysis method:

The project gets revenue from the sale of electricity from the project activity, hence cannot apply simple cost analysis as per Option I. Furthermore, Option II investment comparison analysis cannot be applied as the alternative to the project activity is the electricity generated by new and existing grid connected power plants. Hence the PP has applied the Option III benchmark analysis method to demonstrate the additionality of the project activity in terms of decision-making context which is acceptable to the assessment team. The project cost involves both equity and debt, PP has selected Post tax equity IRR as a financial indicator to demonstrating the financial unattractiveness of the project. Furthermore, the financial indicator selected by the PP is appropriate because the tool does not limit the PP to use either the project IRR or the equity IRR. The PP has the discretion to choose the best indicator based on their preference to know the IRR based on their equity or debt investment. The same is reasonable and acceptable to the assessment team.

Sub-step 2b: Option III. Apply benchmark analysis:

Post tax equity IRR has been chosen as the financial indicator for the demonstration of financial unavailability for the project activity. Since, the PP is demonstrating financial unattractiveness of the project and the project cost involves both equity and debt, post-tax equity IRR is considered to be the appropriate option to indicate financial unattractiveness; and the same is accepted by the assessment team.

As per para 15 of Investment analysis /12/, "The applied benchmark shall be appropriate to the type of IRR calculated. Local commercial lending rate or WACC are appropriate benchmark for a project IRR. Required/expected returns on equity are appropriate benchmark for an equity IRR. Benchmark supplied by relevant national authorities are also appropriate. The Project Verification body shall validate that the benchmark used are applicable to the project activity and the type of IRR calculation presented."

Further para 16 of tool 27/12/ states that "In situation where an investment analysis is carried out in nominal terms and the available IRR benchmark are in real term, project Proponent shall convert the real term values of the benchmark to nominal values by adding the inflation rate. The inflation rate shall be obtained from the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, the target inflation rate of the central bank shall be used. If this information also not available, then the average forecasted inflation rate for the host country published by the IMF (International Monetary Fund World Economic Outlook) or the World Bank for the next five year after the start of the project activity shall be used". The equity IRR calculated is nominal equity IRR. Accordingly, PP converted the default benchmark which is in real terms into nominal terms by using the following equation;

Nominal Benchmark = {(1+Real Benchmark) x (1+Inflation rate)}-1

The assessment team referred the book 'Corporate Finance: Theory and Practice', 2nd edition, by Aswath Damodaran' /47/. In page 320 of the book, the same equation is mentioned for converting real into nominal values. Hence the VCS project verification team considers the above equation as appropriate for converting real benchmark into nominal benchmark.

Parameter	*Project's Specifies*	*VVB Assessment.*
Investment decision date	13/08/2021	The investment decision-making date of the project which is the date of EPC signing. Even though the tender award (24/10/2019) and the PPA signing (30/06/2020) occurred earlier, the PPA contained conditions requiring the project proponent to secure a construction contract, financing agreement, and an executed land lease agreement before the "Effective Date" of the PPA. The PPA explicitly states that if these conditions were not met by the Scheduled Effective Date, either party could terminate the agreement, leading to the cancellation of the project. Given this, the true investment decision was contingent on securing these key project elements. Since the EPC signing was the first binding step that led to the finalization of the land lease and financing—both essential for the PPA's effectiveness—the EPC contract signing date/21/ is considered the investment decision date which is acceptable to the assessment team.

Type of Benchmark	Post-tax equity IRR	In accordance with paragraph 15 of Tool 27: Investment Analysis, version 13.0/27/, it is advised that required or expected returns on equity serve as suitable benchmarks for an equity IRR.
Default Benchmark value	14.15% default for Energy Industries (Group 1) in Cambodia in appendix Tool 27 Investment analysis/12/.	The PP has opted for the default value for Cambodia as outlined in Appendix of EB120, Annex 3, to showcase additionality, which was the most recent default Value as per latest version of Investment Analysis Tool version 13.0 which was the latest version available during the public commenting period of the project activity and submission of documents to VVB. During the investment decision, Tool 27 version 10.0 was the latest available. The default benchmark value as per version 10 is 15.24%. Upon comparison of the default value of 14.15% specified in Appendix of EB 120, Annex 3 with the default value of 15.24% specified in Appendix of EB105, Annex 06, for Cambodia, it was noted that the chosen value appeared conservative. Consequently, it was accepted.
Inflation forecast (Median) 5 years	2.95%/48/	The value has been obtained from the World Economic Outlook database/48/. It is considered suitable since there is no inflation forecast available or target inflation rate published by the central bank of the host country. Therefore, it is deemed appropriate and aligns with the guidelines of Tool 27/12/.
Benchmark value	17.52%	The project proponent has selected the default value for Cambodia in accordance with the Appendix of EB 120, Annex 3, to illustrate additionality, which was the most recent default Value as per latest version of Investment Analysis Tool version 13.0. Additionally, the project proponent possessed a five-year inflation forecast for Cambodia from the IMF World Economic

		Outlook at the time of the investment decision. The assessment team thoroughly examined all of the aforementioned details and documents and affirmed that the identified benchmark for comparing the financial viability of the project activity is appropriate.

The assessment team has verified all the above said documents and confirmed that the benchmark identified to compare the financial attractiveness of the project activity is appropriate.

Sub-step 2c: Calculation and comparison of financial indicators

To ensure accurate calculation of financial indicators, the IRR sheet provided by the PP was thoroughly examined to confirm the inclusion of all pertinent costs and revenues. Additionally, all assumptions and estimates utilized for input values were cross-checked against appropriate sources for validation.

Chronology of events of the project activity are provided below:

Date	Milestone
24 October 2019	Tender Award /35/
13 June 2020	PPA between EDC & Prime Road Alternative (Cambodia) Co., Ltd /16/
19 November 2020	Financial Technical Due Diligence Report by AFRY (Thailand) Limited ("AFRY") /49/
13 August 2021	EPC contract signed between Prime Road Alternative (Cambodia) Co., LTD and China CACS Engineering Corporation /21/ (Investment Decision Date)
30 August 2021	Land Lease Agreement between Prime Road Alternative (Cambodia) Co., LTD and EDC /34/
31 August 2021	PPA effective date declaration by EDC /16/
23 November 2021	Direct Agreement between EDC, Prime Road Alternative (Cambodia) Co., LTD and Bred Bank (Cambodia) PLC /16/
11 November 2022	Project was successfully energized and synchronized /20/

The Equity IRR of the VCS project activity falls below the benchmark, indicating that the VCS project activity lacks financial attractiveness. Below are the key data parameters used for calculating Equity IRR, sourced from the Technical Due Diligence Report prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of the investment decision.

Parameter	Values	Unit	VVB Assessment
Capacity of the project	60 (AC) 77.006 (DC)	MW	Verified against TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and cross verified against the EPC contract/21/, Commissioning certificate/20/, LOA/35/ and PPA/16/ signed between Prime Road Alternative (Cambodia) Co., Ltd and ELECTRICITÉ DU CAMBODGE. Further, the same has been confirmed during onsite visit by the assessment team and found to be correct.
Annual Net generation	141719	MWh	The Annual net generation value is verified against TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 (In-line with the CDM EB48, Annex 11). Additionally, the project verification team has cross-checked the actual electricity generation report/31/. In the first year of operation, the project generated 138,904 MWh (for the year 2023). It was found that the electricity generation in the most recent reports did not exceed the breaching value. The assessment team also cross checked the generation value by multiplying the capacity (DC) with the Specific photovoltaic power output data from the Global Solar Atlas website/40/, i.e., 77.006 MWp * 1555.6 kWh/kWp = 119,805 MWh, which is well within the breaching value. A sensitivity analysis has been conducted the IRR breaches the benchmark value at a electricity generation variation of more than

			250,2766 MWh at which the electricity generation value is 76.60% which is unlikely to happen. Hence, assessment team confirms that the value considered for the project activity is appropriate; hence acceptable.
Project cost	41224.142	k USD	The project cost is sourced from the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision. The total estimated investment cost for project capacity of 60 MW works out to be 41,224.142 k USD. However, as a primary check, the verification team cross checked the actual cost incurred by the project Proponent for the project activity through Common Terms agreement /25/ evidence for the investment. The assessment team confirmed that the actual investment costs of the project is 43.6 Mn USD, which is higher than the estimated value of 41.2 Mn USD TDD/49/. Therefore, assessment team confirmed that the values in TDD/49/ is reasonable and conservative. The assessment team also cross-checked the project cost with the registered VCS project, VCS 4036, titled 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia, which is also commissioned in the year 2022 and found that the per MW project cost was considered as 0.95 million USD/MWp. In contrast, the project cost considered by this project is 0.525 million USD/MWp. Therefore, the per MW considered for this project is found acceptable by the assessment team. The assessment team conducted a sensitivity analysis on the project cost and determined that the IRR would only surpass the benchmark if the project cost were reduced by 48.00%. However, such a reduction is unlikely

			to occur. Consequently, the assessment team has accepted the current project cost.
Debt	70	%	The debt and equity is based on the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision.
Equity	30	%	
Debt	28856.90	k USD	
Equity	12367.24	k USD	The debt-to-equity ratio considered is 70:30% based on the internal assumptions by the project Proponent. Whereas, for a conservative approach PO has considered 69.7:30.3% debt to equity ratio as mentioned in the Common Terms agreement /25/. The equity IRR at this ratio is much below the benchmark. The IRR was cross-checked using various ratios such as 50:50, 80:20, etc., and in all scenarios, it did not surpass the benchmark value. Therefore, the debt equity ratio considered in the investment analysis has been accepted by the VVB.
Interest rate	7	%	The interest rate is based on the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision.
Debt Repayment tenure	17	years	The Debt Repayment tenure is based on the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision.
O&M Cost	807.039	k USD	The value has been sourced from the pre-TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision. This is in line with the methodology tool “Investment analysis” version 13.0/12/. As per TDD, the total Operational Expenditures (OPEX) amounts to 807,039 USD, i.e., 10.3 USD/kWp. The cost break-up of OPEX as per TDD is as below:
Escalation in expense	0.03		

			O&M Cost = 308,000 USD/year, i.e., 3.9 USD/kWp Other non-technical operational costs= 449,039 USD/year O&M Contingency= 50,000 USD/year The assessment team also cross-checked the actual O& M agreement/36/ and found that the yearly fees for O&M service for the 5 years is: Year 1: 409,393.2 USD Year 2: 412,731.17 USD Year 3: 419,806.12 USD Year 4: 427,093.33 USD Year 5: 434,599.15 USD The assessment team also cross-checked the O&M cost with the registered VCS project, VCS 4036, titled 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia, which is also commissioned in the year 2022 and found that the O&M cost was considered as 398,972 USD. A sensitivity analysis has been conducted which shows even if cost of O&M services is reduced by 100% which is very unlikely to happen, the benchmark would not have been breached. Hence, the given values were assessed to be reliable and correct.
1st Year tariff	0.03877	USD/kWh	The value is derived from TDD /49/. Via checking the PPA/16/ and actual sales receipts of electricity/23/ for year 2023, it is confirmed that the Electricity tariff is fixed as 0.03877 USD/kWh which is same to the one used in TDD. And due to the PPA/16/ was signed with the grid which has ensure the Electricity tariff is fixed as 0.03877 USD/kWh, so it is confirmed that the project participant has use the highest possible tariff that they may receive by supplying the electricity to the

			grid to assess the economic attractiveness of the project activity which is verified in line with the request of section 34 of the applied methodology. If the Electricity tariff increases by 76.60%, the IRR of project activity reach the benchmark. Based on above assessment, it is impossible to increase Electricity tariff by 76.60% to make the IRR to cross the benchmark. The assessment team also cross-checked the Tariff of the registered VCS project, VCS 4036, titled 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia, which is also commissioned in the year 2022 and found that the electricity tariff (without VAT) is considered as 0.076 USD/kWh. Hence the tariff considered for the project activity is conservative and acceptable to the verification team.
Lifetime of the plant	20	Years	The lifetime of the project is defined as 20 years in TDD/49/ and cross checked with PPA/16/. the considered duration is correct and consistent.
SLM Depreciation Rate (Book)	5	%	The salvage value of the equipment is almost equal the decommissioning cost and also the land is a property of EDC/34/ and hence the land value is not considered as part of salvage value. The depreciation rate was cross checked with the Cambodian 2018 Tax Booklet/39/ and found that the value considered by the PP is correct and consistent. The assessment team also cross-checked with the registered VCS project, VCS 4036, titled 26MWp Solar Photovoltaic Power Plant in Bavet City, Cambodia, which is also commissioned in the year 2022 and found that the depreciation rate is considered as

			5%. Thus, the value considered for the project activity is acceptable to the verification team.
Tax holiday	9	Years	The Tax holiday is based on the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision. Further, assessment team also checked the publicly available source, i.e., incentives and schemes published by Council for the development of Cambodia/55/ (https://cdc.gov.kh/incentives-and-schemes/) and found that the value considered by the PP is correct and consistent. Thus, it is acceptable to the assessment team.
Income tax rate from 10th year onwards	20	%	The tax rate is cross checked and found to be correct which was applicable at the time of investment decision. Further, assessment team also checked the publicly available source, i.e., standard rate of corporate income tax (CIT) published by PwC Cambodia/54/ and found that the value considered by the PP is correct and consistent. Thus, it is acceptable to the assessment team.
Annual Degradation Factor	0.45	%	The Annual Degradation factor is based on the TDD/49/ which is prepared by third party AFRY (Thailand) Limited, dated 19/11/2020 and available at the time of investment decision. The same was also cross-checked with the technical specifications of PV module from the manufacturers' datasheet/22/. Hence, assessment team confirms that the value considered for the project activity is appropriate and hence acceptable. Further, generation values have also subjected to sensitivity analysis. The assessment team also cross-checked with the registered VCS project, VCS 4036, titled 26MWp Solar Photovoltaic Power Plant in

			Bavet City, Cambodia, which is also commissioned in the year 2022 and found that the degradation rate is considered as 0.5%. Thus, the value considered for the project activity is acceptable to the verification team.

Financial calculation and conclusion

The post-tax equity IRR calculations were provided in a spreadsheet. The calculation was verified and found to be correct by project verification team; as well as the assumptions used in the calculation were deemed to be correct. The post-tax equity IRR without carbon credit revenues is 1.31% which confirms that the project activity in absence of the carbon credit benefits and compared to the benchmark return on equity 17.52% is not financially attractive.

Sub Step 2d: Sensitivity Analysis:

The Guidance on Assessment of Investment Analysis requires the robustness of the conclusion arrived at to be proved through a sensitivity analysis by varying the critical assumptions to a reasonable variation. The project developer has identified generation, project cost, O&M cost, tariff as critical assumptions. These constitute more than 20% of the project cost/revenue. Guidance 28 of Tool 27 states that as a general point of departure, variations in the sensitivity analysis should at least cover a range of +10% and –10%, unless this is not deemed appropriate in the context of the specific project circumstances.

Variation %	-10%	Normal	10%	Variation required to reach benchmark	Value required to reach benchmark	
Tariff	-1.23%	1.31%	3.68%	76.60%	0.068	USD/kWh
Generation	-1.23%	1.31%	3.68%	76.60%	250276	MWh
Project Cost	3.44%	1.31%	-0.51%	-48.00%	21436.55	k USD
O&M Cost	1.79%	1.31%	0.82%	NA*	NA*	k USD

*Even with the zero O&M cost the IRR will not reach benchmark. Hence no value is provided

The results of sensitivity analysis show that even with a variation of ±10% in tariff, Generation, project cost, and O&M cost, equity IRR is significantly lower than the benchmark. And it is evident from the results given above; the project remains additional even under the most favorable conditions.

Generation: Project is already operational and the actual generation is less than the generation used in the IRR calculation. The annual actual electricity generation achieved since the commissioning is less than the value required to breach the benchmark even with the generation of 138,779 MWh achieved in the year 2023.

IRR will only cross the benchmark if the generation increased more than 76.60%, Hence, there is no possibility of a further increase to generation at the rate 76.60%.

O&M Cost: The O&M agreement/36/ is already in place by the project Proponent and O&M used in the calculation is higher than the actual O&M. sensitivity analysis reveals that O&M will breach the benchmark at negative values and is hypothetical case. Hence, there is no possibility of further decrease and is highly unlikely.

Project Cost: The investment analysis considers a project cost of $41.2 million USD, derived from negotiations with the supplier as per the TDD/49/. The Internal Rate of Return (IRR) would surpass the benchmark if there were a variation in project cost of -48.00%, necessitating an additional investment of $43.6 million USD. However, this scenario is not feasible as the project has already been commissioned at the estimated cost outlined in the IRR, as evidenced by the Common Term Agreement/25/. Hence, there is no possibility of further decrease in the project cost at the rate 48.00%.

Tariff: As per the LOA/35/ and Power Purchase agreement/16/ the tariff rate of electricity is 0.03877 USD/kWh the same is consistent with value in the TDD/49/ which is taken for Investment analysis. The IRR will only cross the benchmark at the variation of 76.60% as per the PPA the tariff is fixed and there is not any chances to further variation. However, the actual tariff based on the PPA/16/ signed is same as the tariff considered in the IRR calculation i.e., 0.03877 USD/kWh which is fixed for PPA tenure (20 years) without any escalation which is much below the tariff value required to breach the benchmark value. Hence acceptable by the assessment team.

All the four scenarios highly hypothetical and impossible. Verification Team has arrived at the conclusion that the project scenario is not economically feasible without benefits from carbon benefits.

Step 3: Barrier Analysis

The additionality of the project has been demonstrated by applying the investment analysis, thus no barrier analysis is carried out.

Step 4: Common Practice Analysis

The section below provides the analysis as per step 4 of the "Tool for the demonstration and assessment of additionality", version 7.0.0/11/ and according to "Common Practice" Tool version 03.1/13/.

Step 1: Calculate applicable capacity or output range as +/- 50% of the total design capacity or output of the project activity:

The project installed capacity is 60 MW. Therefore, total capacity of power plants which will be included in the analysis will be between 30 MW to 90 MW.

Step 2: Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- The projects are located in the applicable geographical area;

The project is located in Kampong Chhnang province, Cambodia and the applicable geographical area is Cambodia. All the projects in the host country Cambodia have been chosen for analysis.

- The projects apply the same measure as the project activity;

Renewable Energy Projects

- The projects use the same energy source/fuel and feedstock as the project activity, if a technology switch measure is implemented by the project activity;

Solar power projects

- The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the project plant;

The project activity produces electricity; therefore, all solar power plants that produce electricity are candidates for similar projects;

- The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;

Range in between 30 MW to 90 MW

As per the Tool24: "Common practice", Version 3.1/13/ and the CDM Glossary, the start date i.e. the EPC contract signing date for the solar power plant is 13/08/2021. Therefore, projects which have started commercial operation between 04/07/2002 to 13/08/2021 have been considered for analysis.

VVB team has cross-checked with the publicly available sources, numbers of Similar projects (CDM and non-CDM) identified based on available published sources and for project activity 5 have started commercial operation between 25/09/2002 to 14/06/2018 have been considered for analysis. which fulfils the above-mentioned conditions are 5 i.e. Nsolar = 5

Project Name	**Capacity (MW)**	**COD**	**Registered Activities**
Risen Energy Battambang Solar Park	60	March 31st, 2021	I-REC
Pursat Solar	90	Mar-21	
Kampong Speu Solar	80	Dec-19	

Kampong Chnnang	60	Nov-20	
Ray Power – 30 MW Solar Photovoltaic Power Plant in Banteay Meanchey Province	30	December 15th,2020	I-REC

Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing VVB team. Note their number, Nall.

Within the 5 projects identified in Step 2, 2 projects are registered under the International Renewable Energy Certificates (I-RECs) mechanism.

Nall = 3

Step 4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the project activity. Note their number Ndiff.

Projects with technologies different to technology applied in the proposed project activity were identified as Ndiff = 0.

Step 5: calculate factor F= 1 – (Ndiff/Nall) representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the project activity that deliver the same output or capacity as the project activity.

The factor F was found to be in line with Tool 24

F = 1-Ndiff/Nall

F = 1-(0/1) = 1

Nall - Ndiff = 1 - 0 = 1

Outcome of Step 5:

As,

- F = 1 which is greater than 0.2
- Nall-Ndiff = 1; is not more than 3

Since Nall-Ndiff is equal to 1, the project activity is “not a common practice” within a sector in the applicable geographical area.

Conclusion:

As described above, the project fulfils all necessary requirements of additionality specified in the 'Tool for the demonstration and assessment of additionality' v7.0.0/11/. Hence, the project is additional.

3.4.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

The emission reduction (ERy) by the project activity during the crediting period is the difference between baseline emissions (BEy) and project emissions (PEy). All aspects related to the direct and indirect GHG emissions as relevant to the project activity have been addressed and are presented in a transparent manner, in line with the applied methodology ACM0002, version 21.0/9/.

The assessment team checked whether the equations and parameters used to calculate GHG emission reductions or net anthropogenic GHG removals are in accordance with applied methodology. Verification team checked section 5 of the joint PDMR/01/ to confirm whether all formulae used to calculate baseline emissions, project emission and leakage have been applied in line with the underlying methodology.

Baseline Emissions:

The baseline emissions as discussed in section 5.1 of the PDMR/01/ mentioned that the emission would have occurred in the absence of the project activity. The emission reduction calculation has been done as per the Large-scale Consolidated Methodology- ACM0002 "Grid-connected electricity generation from renewable sources", Version 21.0/09/.

The baseline emissions of the project activity according to the paragraph 47 of the applied methodology is,

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO2/yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO2 emission factor for grid connected power generation in year y calculated using TOOL07 (t CO2/MWh)

As per paragraph 49 of the applied methodology, If the project activity is the installation of a greenfield power plant

EGPJ,y = EGfacility,y

Where EGfacility,y = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/year)

As per JPDMR the estimated net electricity generation from the project activity is 139,279 MWh (annual average over the crediting period) and calculated combined margin emission factor based on the Tool is 0.5878 tCO2e/MWh. Hence the baseline emission value will be 81,868 tCO2e (annual average over the crediting period).

The electricity generation for the project activity is sourced from the Technical Due Diligence report/49/ prepared by third-party, AFRY (Thailand) Limited. PLF is calculated using the generation data (P90), given in the report, i.e., 141719/ (60*24*365) = 26.96%. Hence the value considered by the project Proponent to arrive at the ex-ante emission reductions of the project is deemed acceptable to the verification team and also in line with paragraph 3 (b) of "Guidelines for the reporting and Validation of Plant Load Factors" (Annex 11 of EB 48)/49/. Hence, the value considered for the calculation of emission reductions for the project activity is reasonable and appropriate. For ex-post, this parameter ($EG_{facility,y}$) is being calculated as difference of electricity exported to the grid by the project activity ($EG_{export,y}$) and electricity imported from the grid by the project activity ($EG_{import,y}$) and those are being measured by energy meters of accuracy class 0.5s (main meter) and 0.5s (check meter).

Project emissions:

As per paragraph 35 of the applied methodology/9/, For most renewable energy project activities, PEy = 0. Since solar power is a GHG emission free source of energy project emission considered as Zero for the project activity.

Leakage Emissions:

As per the paragraph 61 of the applied methodology/9/, there are no emissions related to leakage in this project.

Emission reductions:

As per Paragraph 62 of the applied methodology/9/, emission reductions are calculated as follows

$ER_y = BE_y - PE_y - LE_y$

Where:

ERy = Emission reductions in year y (tCO2e/y)

BEy = Baseline Emissions in year y (t CO2/y)

PEy = Project emissions in year y (t CO2/y)

LEy = Leakage emissions in year y (t CO2/y)

Hence the annual emission reductions based on the ex-ante parameters is 81,868 tCO2e/2/ (Annual Average over the crediting period).

3.4.7 Methodology Deviations

The project has applied a deviation from the employed methodology, ACM0002 (Version 21.0)/9/ for calculating the baseline emission factor (EF). According to this methodology, the baseline emission factor for the project should be the combined margin emission factor ($EF_{grid, CM, y}$) for the grid, which is calculated as a combination of the operating margin emission factor ($EF_{grid, OM, y}$) and the build margin emission factor ($EF_{grid, BM, y}$). The calculations should follow the steps defined in the "Tool to Calculate the Emission Factor for an Electricity System" (Version 7.0).

Upon reviewing the publicly available data, including information from the EDC grid website/41/, it was confirmed that necessary data for calculating the operating margin and build margin factors were unavailable. Specifically, fuel consumption data for the EDC grid over the last five years and CO2 emission factors for individual plants were not provided. The default grid emission factors published by Cambodia's Ministry of Environment are based on data from 2010, 2011, and 2012/38/. These data gaps prevent the calculation of the combined margin emission factor ($EF_{grid, CM, y}$) in accordance with the prescribed methodological tool.

Further review showed that there is no available updated emission factor for the EDC grid that has been approved by the relevant Designated National Authority in Cambodia, as confirmed by checking the DNA website/42/.

Given the unavailability of relevant data to calculate the $EF_{grid, CM, y}$ the project adopted the Default Emissions Factor (DEF) issued by the IFI Technical Working Group (IFI TWG). This DEF is used as a substitute for the grid's emission factor.

The most recent version of the IFI Dataset of Default Grid Factors v.3.2, published in April 2022/43/, was identified as the latest available source prior to validation. This DEF complies with the requirements of CDM Tool 07, "Tool to calculate the emission factor for an electricity system" (Version 07.0) /10/, and was verified as accurate.

According to the IFI Technical Working Group on GHG accounting, the DEF dataset provides a Combined Margin (CM) emission factor, which is derived by combining an Operating Margin (OM) and a Build Margin (BM), as required by the methodology (CDM Tool 07). The data used in the DEF calculation is sourced from the International Energy Agency's (IEA) energy statistics database, further verifying its credibility and consistency with international standards.

Based on the aforementioned source, it has been verified that the latest $EF_{grid, CM, y}$ for Intermittent energy (e.g., solar, wind, tidal) is 0.8743 tCO_2e/MWh and Firm energy (e.g., hydro, geothermal) is 0.5878 tCO_2e/MWh in Cambodia for 2021. The Project Participant (PP) has selected an emission factor of 0.5878 tCO_2e/MWh to calculate the emission reductions for the project as a conservative approach.

Based on the assessment and verification of available data sources, it is concluded that the Default Grid Emissions Factor (DEF) from the IFI Technical Working Group is the most appropriate and conservative emission factor for the project. This factor aligns with the requirements of CDM

Tool 07 and is based on credible data sources, including the IEA energy statistics and the IFI grid factor dataset.

Therefore, the methodology deviation is justified, and the emission factor used for the project is both reasonable and conservative.

3.4.8 Monitoring Plan

The monitoring plan of the project activity is in compliance with the monitoring methodology ACM0002: "Grid-connected electricity generation from renewable sources", version 21.0/9/. The assessment team, based on the onsite inspection conducted, confirms that the project proponent is monitoring the project activity as per the monitoring plan included in section 6 of the joint PDMR/01/.

Parameters determined ex-ante:

The project adopts the ex-ante calculation for the following parameters:

Parameter	Unit	Source/Justification
$EF_{grid,CM,y}$ - Combined margin emission factor of the grid	tCO_2/ MWh	During the 1st crediting period, the value for $EF_{grid, CM, y}$ is considered as 0.5858 tCO_2/ MWh. Due to the unavailability of specific fuel consumption data for Cambodia's national power grid for the past five years, along with the lack of an updated grid emission factor approved by the Designated National Authority (DNA) of Cambodia, PP has considered the data from the most recent Harmonized IFI Default Grid Factors 2021 (Version 3.2). Based on the aforementioned source, it has been verified that the latest $EF_{grid, CM, y}$ for the Intermittent energy (e.g., solar, wind, tidal) is 0.8743 tCO_2e/MWh and Firm energy (e.g., hydro, geothermal) is 0.5878 tCO_2e/MWh in Cambodia for 2021. This value is reported ex-ante and will remain the same for the entire crediting period which is in-line with the VCS PDMR/01/. Thus, the value considered by the project proponent is appropriate and in relation to the project activity and its acceptable to the assessment team.

Parameters monitored ex-post:

According to the methodology ACM0002, version 21.0/9/ following parameters are monitored.

Parameter	Unit	Source/Justification

<table>
<tr>
<td>$EG_{facility,y}$: Quantity of net electricity displaced in year y in MWh/y</td>
<td>MWh/year</td>
<td>Value applied: 141,719 MWh/year

The monitoring parameter will be continuously tracked using energy meters (Main and Check Meters). These metering devices are supplied and installed by the EDC, with all billing meters for the incoming feeders having an accuracy class of 0.5s. These meters are located at the interconnection/substation at the power plant premise.

The net electricity supplied by the project is determined by the difference between the exported and imported electricity. Export and import readings will be obtained from the Joint Meter Reading (JMR) report.

The EDC has approved, tested, and sealed the meters, retaining custody of them, and ensures they are calibrated annually. The monthly electricity supplied or exported by the project, as documented in the JMR report/50/, is cross-verified against the monthly sales invoices/23/. In the event of any delay or absence of calibration, appropriate procedures will be followed to maintain conservative metering. The EDC is responsible for the metering setup, including the accuracy class of the meters and the frequency of calibration, and the project proponent (PP) has no authority over these aspects. The PP receives data on the net electricity supplied to the grid, as provided by the EDC, which serves as the monitoring parameter. Billing is based on the substation meters.

Electricity meters, the details of which are as below:
<table>
<tr><th>Feeder</th><th>Main Meter Number</th><th>Backup Meter Number</th><th>Accuracy</th><th>Make</th></tr>
<tr><td>1</td><td>55548513</td><td>55548515</td><td rowspan="4">0.5s</td><td rowspan="4">Landis+Gyr E650</td></tr>
<tr><td>2</td><td>55548514</td><td>55580597</td></tr>
<tr><td>6</td><td>55548500</td><td>55580600</td></tr>
<tr><td>7</td><td>55548498</td><td>55548501</td></tr>
</table>
</td>
</tr>
</table>

		Based on the meter reading statement (JMR) to Project Proponent, invoices will be raised. These invoices can be used for cross checking the meter readings taken for the respective project activity. All data collected as part of monitoring will be archived electronically and be kept at least for 2 years after the end of the crediting period or till the last issuance of ACCs for the project activity whichever occurs later. During the physical site visit, the assessment team found that the monitoring practices followed by the project proponent is appropriate and in relation to the project activity and its acceptable to the assessment team.

The responsibilities and authorities for project management, procedures for monitoring and reporting, and QA/QC procedures have been systematically established and formalized.

Details of the data collection and frequency of data recording and associated formats are described and observed to be adequate. The monitoring plan will give opportunity for real measurements of emission reductions that will be accrued.

The monitoring plan and the QA/QC procedure would ensure the accurate monitoring of the data/parameters mentioned in the VCS JPDMR throughout the implementation of the project activity. The assessment team confirms that the monitoring procedures mentioned are best as per Industry norms.

Opinion: The assessment team confirms that:

(a) The monitoring plan mentioned in VCS JPDMR/1/ is in compliance with the applied methodology ACM0002 version 21.0/9/;

(b) Based on the on-site interview and document review, the assessment team confirms that monitoring arrangements described in the monitoring plan are feasible within the project design;

(c) Based on the on-site interview and document review, the assessment team confirms that project participant deputed the competent personal to execute the monitoring approach and to follow the monitoring plan.

3.5 Non-Permanence Risk Analysis

The project activity does not require a non-permanence risk analysis. Hence this is not applicable.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

Implementation Status	Assessment steps, evidence checked, & conclusion:
Project implementation	The project has been implemented as per the project description provided in the section 4.1 of the joint PD&MR. There are no material discrepancies between project implementation and the project description provided in the PDMR/01/.
Monitoring plan	The monitoring plan has been found to be completed and implemented at the location where the project is located. The implemented monitoring system was found suitable with respect to the process and schedule for obtaining, recording, compiling and analysing the monitored data and parameters. No discrepancies between the actual monitoring system, and the monitoring plan set out in the project description and the applied methodology have been found during the site visit.
AFOLU-specific project implementation	Not Applicable as the project activity is not an AFOLU Project.

Opinion:

The assessment team concludes the following:

- There is no material discrepancies between project implementation and the project description provided in the PDMR/01/.
- The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters) is appropriate.
- There is no material discrepancies between the actual monitoring system and the monitoring plan set out in the PDMR/01/ as per the applied methodology.
- The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

- The project has not participated or been rejected under any other GHG programs. Further the project has not received or sought any other form of environmental credit, or has become eligible to do during the validation or verification.
- The project is registered under VCS only. Hence the assessment team confirms that there is no double counting associated with project activity being participation of other GHG programs which is also verified from the no-double-counting declaration given by the PP/28/.

Hence it can be concluded based on the review of the project documents and the site visit that project has been implemented as described in the joint VCS joint PD&MR/1/.

4.2 Accuracy of Reduction and Removal Calculations

The assessment team was able to confirm that the monitoring plan described in PDMR/1/ is in accordance with the applied CDM methodology for the project activity i.e., ACM0002 – Grid-connected electricity generation from renewable sources, Version 21.0/9/ and its applicable tools.

The parameter stated in the monitoring plan/01/ and the applied methodology /9/ have been fulfilled in the current monitoring period.

The GHG emission reductions were correctly calculated on the basis of the applied methodology ACM0002 – Grid-connected electricity generation from renewable sources – Version 21.0/9/ and the formulae given in the JPDMR/1/.

Parameter monitored:

The assessment team confirms through physical site visit and from the document review, the actual monitoring system complies with the monitoring plan mentioned in the PDMR/01/. According to the monitoring plan in the PDMR/01/, the following monitoring parameter is required to be monitored.

Parameter	Unit	Source/Justification
$EG_{facility,y}$: Quantity of net electricity displaced in year y in MWh/y	MWh	**Verified value:** 1. 156,367 MWh (monitored value) 2. 156,242 MWh (after applying error, due to delayed calibration) The electricity supplied or exported by the project, is checked from the JMR report/50/ and is cross-verified against the monthly sales invoices/23/. The calibration details of the energy meters are given below:

Feeder	Main Meter Number	Backup Meter Number	Commissioning date/Calibration date	Validity
1	55548513	55548515	11/11/2022	10/11/2023
2	55548514	55580597		
6	55548500	55580600		
7	55548498	55548501		
1	55548513	55548515	20/03/2024	19/03/2025
2	55548514	55580597	21/03/2024	20/03/2025
6	55548500	55580600	22/03/2024	21/03/2025
7	55548498	55548501	22/03/2024	21/03/2025

From the review of calibration records by assessment team, it is found that the calibration was delayed for the period from 10/11/2023 to 31/12/2023 during the monitoring period. Hence, the maximum possible error of 0.5% is applied for the delayed period as below:

Export = Measured value * (1-0.5%)

Import = Measured value *(1+0.5%)

The monitoring parameter, Unit and description, source, measurement methods and procedures, frequency of monitoring are in accordance with the JPDMR/01/. During the physical site visit, the assessment team found that the monitoring practices followed by the project proponent is appropriate and in relation to the project activity and its acceptable to the assessment team.

4.3 Quality of Evidence to Determine Reductions and Removals

All relevant documents were checked to assess the correctness and quality of data submitted by the project proponents, which are used to determine emission reductions.

All records needed for monitoring are archived in line with the requirements of the monitoring plan. No significant lack of evidence and missing data were detected during onsite audit discussion and inspection and the document review. Hence, the assessment team confirms that the monitoring system ensures required quality of the monitoring system to ensure the quality of the monitored data. All internal data are subjected to QA/QC measures.

The assessment team can confirm that sufficient evidence is available for the whole monitoring period and the same is verifiable and that the data collection system meets the requirements of the monitoring plan and the applied methodology according to the assessment carried out on site and in the document review.

Assessment team confirms that the quality of evidence to determine the GHG reductions produced was found satisfactory. The detailed information flow with the roles and responsibilities of the individuals and the monitoring system have been provided in the JPDMR/01/.

It was also verified through onsite audit inspection that the team involved in the monitoring of project activity is well experienced. Hence, the verification team concludes that competent staff is employed by the project proponent to carry out the relevant tasks with sufficient accuracy.

5 VALIDATION AND VERIFICATION OPINION

5.1 Validation and Verification Summary

4K Earth Science Private Limited (4KES) has been contracted by 'Kosher Climate India Private Limited' to performed the independent joint validation and verification of the emission reductions for the VCS project activity (VCS ID - 4869) "Kampong Chhnang 60 MW Solar Power Project" in Cambodia, with regard to the relevant requirements of VCS Standard version 4.7; to attain real, measurable, additional and permanent emission reductions.

4KES commenced the validation and verification on the basis of the baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources, version 21.0/9/, the monitoring plan contained in the Joint PDMR /01/ (Version 3.0 dated 20/02/2025) as per the process described under Section 2 of this report.

The validation service is for the first crediting period of 7 years from 11-November-2022 to 10 November-2029 and verification for the monitoring period 11-November-2022 to 31-December-

2023 as reported in the Joint Project Description and Monitoring Report Version 3.0 dated 20/02/2025. The project proponent correctly applied the baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources, version 21.0/9/. The assessment team confirms that the validation and verification of the GHG statement was conducted in accordance with ISO 14064-3: 2019.

The project activity involves installation of 60 MW (AC) Solar Power Plant by Prime Road Alternative (Cambodia) Co., Ltd. The project involves deploying Solar Photovoltaic panels to harness solar radiation, converting it into DC power, and subsequently utilizing inverters to transform the DC power into AC power. The objective of the project is to produce clean electricity using solar energy and channel this generated power into the National Grid of Cambodia via the electricity transmission line through power purchase agreement, there by displacing electricity generated by fossil fuel-based power plants from the National Grid of Cambodia. The project is expected to contribute 3 SDGs which are SDG 7, 8 and 13.

During the first crediting period, the project is expected to achieve an annual emission reduction of 81,868 tCO_2e over the 7-year crediting period and a total emission reduction of 573,078 tCO_2e during the first 7-year renewable crediting period/2/.

During the current verification period, i.e., 11/11/2022 to 31/12/2023, the project has achieved an emission reduction of 91,839 tCO_2e/3/.

A risk-based approach has been followed to perform this joint validation and verification. In the course of the validation and verification, 20 Corrective Action Request (CAR) and 15 Clarification Request (CL) were raised and successfully closed. 0 FARs raised during the joint validation and verification.

The joint validation and verification is based on the VCS JPDMR/01/, Version 3.0 (20/02/2025), additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and supporting documents made available to the assessment team by the project proponent.

As a result of the validation and verification, the assessment team confirms that:

- The project fulfils criteria of VCS Standard Version 4.7/6/.
- The project is in line with all relevant VCS requirements.
- The project baseline is sufficiently justified as per the applied methodology and Tools /9/ to /13/
- All the data and information used for ex -ante calculation of emission reductions is correctly applied in the VCS JPDMR/01/
- A reasonable level of assurance has been applied
- No restrictions or uncertainties were identified related to the validation and verification.
- Project has been implemented and operated as per VCS JPDMR/01/

4KES's validation and verification approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. 4KES planned and performed the validation and verification by obtaining evidence and other information and explanations that 4KES considered necessary to give reasonable assurance that reported GHG emission reductions are fairly stated.

5.2 Validation Conclusion

In our opinion, the GHG emissions reductions estimated for the project activity for the period 11-November-2022 to 10 November-2029 are fairly stated in the Joint Project Description and Monitoring Report/01/, Version 3.0 dated 20/02/2025. The GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources, version 21.0/9/ and the VCS standard v4.7 /06/. The assessment team confirm that validation of the GHG statement was conducted in accordance with ISO 14064-3: 2019.

The validation is based on the Joint PDMR/01/, additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and supporting documents made available to the assessment team by project proponent.

The review of the project design and the subsequent follow-up interviews have provided 4KES with sufficient evidence to determine the project's fulfilment of all the above stated criteria. No restrictions or uncertainties were identified related to the validation.

Therefore, 4KES is able to certify that the project is in full compliance with the VCS Standard Version 4.7/06/ and recommend registration of the project activity.

Crediting Period: From 11-November-2022 to 10 November-2029

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
11-Nov-2022 to 31-Dec-2022	11,639	0	0	11,639	NA	11,639
01-Jan-2023 to 31-Dec-2023	82,928	0	0	82,928	NA	82,928
01-Jan-2024 to 31-Dec-2024	82,555	0	0	82,555	NA	82,555

01-Jan-2025 to 31-Dec-2025	82,183	0	0	82,183	NA	82,183
01-Jan-2026 to 31-Dec-2026	81,813	0	0	81,813	NA	81,813
01-Jan-2027 to 31-Dec-2027	81,445	0	0	81,445	NA	81,445
01-Jan-2028 to 31-Dec-2028	81,079	0	0	81,079	NA	81,079
01-Jan-2029 to 10-Nov-2029	69,436	0	0	69,436	NA	69,436
Total	**573,078**	**0**	**0**	**573,078**	**NA**	**573,078**

5.3 Verification conclusion

The verification has been done with a reasonable level of assurance. In our opinion, the GHG emissions reductions reported for the project activity for the period 11-November-2022 to 31-December-2023 are fairly stated in the Joint Project Description and Monitoring Report, version 3.0 dated 20/02/2025. The GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology ACM0002: Grid-connected electricity generation from renewable sources, Version 21.0/9/ and the VCS standard v4.7/06/. Assessment team confirm that verification of the GHG statement was conducted in accordance with ISO 14064-3: 2019.

Verification Period: 11-November-2022 to 31-December-2023 (both start date and end date included)

Verified GHG emission reductions and carbon dioxide removals in the above verification period:

Vintage period	Baseline emissions (tCO_2e)	Project emissions (tCO_2e)	Leakage emissions (tCO_2e)	Reduction VCUs (tCO_2e)	Removal VCUs (tCO_2e)	Total VCUs (tCO_2e)
11-Nov-2022 to 31-Dec-2022	10,264	0	0	10,264	NA	10,264

01-Jan-2023 to 31-Dec-2023	81,575	0	0	81,575	NA	81,575
Total	**91,839**	**0**	**0**	**91,839**	**NA**	**91,839**

5.4 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
11-Nov-2022 to 31-Dec-2022	11,639	10,264	-11.81%	The variation in the ERs is due to fluctuations in solar irradiance, which directly affects the project's electrical output. As lower irradiance levels reduce power generation, this naturally impacts the emission reductions achieved. As the achieved emission reductions for this solar project activity during the current monitoring period is lower than the estimated emission reductions, it is acceptable to the assessment team.
01-Jan-2023 to 31-Dec-2023	82,928	81,575	-1.63%	
11-Nov-2022 to 31-Dec-2023	94,567	91,839	-2.88%	
Total	94,567	91,839	-2.88%-	

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No commercially sensitive information included in the project description is to be excluded in the public version.

Section	Information	Justification	Assessment method conclusion
Not applicable	Not applicable	Not applicable	Not applicable

APPENDIX 2: REFERENCE DOCUMENTS

The following table lists the documentation that were reviewed during this joint validation and verification of the project activity:

1.	Joint Project Description & Monitoring Report, version 1.0 dated 20/02/2024 (Initial version) Joint Project Description & Monitoring Report, version 2.0 dated 07/11/2024 Joint Project Description & Monitoring Report, version 3.0 dated 20/02/2025 (Final Version)
2	Emission Reduction Calculation Excel sheet (Ex Ante) Version 1.0 dated 10/01/2024 Emission Reduction Calculation Excel sheet (Ex Ante) Version 2.0 dated 07/11/2024
3	Emission Reduction Calculation Excel sheet (Ex post) Version 1.0 dated 10/01/2024 Emission Reduction Calculation Excel sheet (Ex post) Version 2.0 dated 07/11/2024
4	VCS Program Definitions, version 4.5 dated 16/04/2024 https://verra.org/wp-content/uploads/2024/04/VCS-Program-Definitions-v4.5-FINAL-4.15.24.pdf
5	VCS Registration & Issuance Process, version v4.5, dated 16/04/2024 https://verra.org/wp-content/uploads/2024/04/Registration-and-Issuance-Process-v4.5-FINAL-4.15.2024.pdf
6	VCS: VCS Standard, version 4.7 dated 16/04/2024 https://verra.org/wp-content/uploads/2024/04/VCS-Standard-v4.7-FINAL-4.15.24.pdf
7	VCS: VCS Program Guide, version 4.4 dated 29/08/2023 https://verra.org/wp-content/uploads/2023/08/VCS-Program-Guide-v4.4.pdf
8	VCS Validation and Verification Manual version 3.2 dated 19/10/2016
9	CDM approved methodology: ACM0002, "Grid-connected electricity generation from renewable sources", Version 21.0 ACM0002 Version 21.0
10	Tool 07 - Tool to calculate the emission factor for an electricity system, Version 7.0 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf
11	Tool 01- Tool for the demonstration and assessment of additionality, Version 7.0.0

	https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf
12	Tool 27- Investment analysis, Version 13.0 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v14.0.pdf
13	Tool 24- Common Practice, Version 03.1 https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf
14	CDM validation and verification standard for project activities, version 3.0 CDM VVS, Ver 3.0
15	CDM Project Standard for project activities, version 3.0 CDM PS, Ver 3.0
16	**Power Purchase Agreement (PPA):** • Power Purchase Agreement (PPA) between Electricite Du Cambodge and Prime Road Alternative (Cambodia) Co., LTD dated 30/06/2020 • PPA Direct Agreement between Electricite Du Cambodge, Prime Road Alternative (Cambodia) Co., LTD and Bred Bank (Cambodia) PLC dated 23/11/2021 Letter of declaration of PPA effective date by Electricite Du Cambodge (EDC) dated 31/08/2021
17	Google Earth Pro/ Maps
18	VCS Joint Project Description and Monitoring Report Template, version 4.3 https://verra.org/documents/vcs-joint-project-description-monitoring-report-template-v4-3/
19	VCS Joint Validation and Verification Report Template, version 4.4 https://verra.org/documents/vcs-joint-validation-and-verification-template-v4-4/
20	COD: Energization And Commissioning Tests of 60 MW Solar Photovoltaic Power Plant in Kampong Chhnang Province, Cambodia: Energized and synchronized on 11/11/2022. Commissioning tests of the Project from 24/11/2022 to 30/11/2022.
21	Engineering, Procurement and Construction (EPC) Contract between Prime Road Alternative (Cambodia) Co., LTD and China CACS Engineering Corporation dated 13/08/2021.
22	Technical Specification of the Major Equipments installed at site

23	Invoice raised by Prime Road Alternative (Cambodia) Co., LTD to Electricite Du Cambodge dated from November 2022 to December 2023.
24	Calibration records of the measuring instruments used in the project activity. Calibration of Certificate issued by Kingdom of Cambodia Nation Religion King dated March 20th to 23rd 2024.
25	Common Terms Agreement dated 25th August 2021
26	EIA approval issued by Ministry of Environment dated 01/12/2021
27	ESIA report dated March 2021
28	Declaration made by the project proponent for the project activity not applying /seeking carbon credits under other GHG programs
29	Other GHG program registries checked for Double Counting. • https://www.climateactionreserve.org/ • https://registry.goldstandard.org/projects?q=&page=1 • https://iceland.itmoregistry.net/Public/Project • https://biocarbonregistry.com/en/projects/ • https://projects.globalcarboncouncil.com/pages/submitted_projects
30	Employment Records – HR Records, Declaration of employment, HR Policy etc
31	Meter Reading Monthly Report.
32	**LSC:** • Stakeholder Engagement Report is prepared by China CACS & PRAC dated on July 21, 2022. • Stakeholder Engagement Report is prepared by China CACS & PRAC dated on February 24, 2023.
33	Training Records
34	Land Lease Agreement between Electricite Du Cambodge and Prime Road Alternative (Cambodia) Co., LTD dated 30/08/2021
35	Tender - Ownership Document letter of award (LoA) dated October 24, 2019.
36	**O & M Contract:**

	• O & M Contract is issued by China CACS Engineering Corporation to the employer Prime Road Alternative (Cambodia) Co., LTD for the project activity dated 07/09/2021. 6) • O & M Contract is issued by Uper Energy Singapore PTE. LTD. to the employer Prime Road Alternative (Cambodia) Co., LTD for the project activity 31/08/2021.
37	**HR Policies:** • Anti-sexual Harassment policy dated 08/06/2021. • CACS Anti-Sexual Harassment Policy • Prime Anti-Sexual Harassment Policy • CACS Human Resource Policy and Training Plan dated 19/03/2021
38	Default grid emission factor published by Cambodia's Ministry of Environment are based on data from 2010, 2011 and 2012 https://www.iges.or.jp/en/publication_documents/pub/data/en/5097/GEF-Cambodia_2010-2012.pdf
39	Cambodian 2018 Tax Booklet https://www.pwc.com/kh/en/tax/cambodian-2018-tax-booklet1.pdf
40	Global Solar Atlas website https://globalsolaratlas.info/map?c=11.791944,104.397222,11&s=11.791944,104.397222&m=site
41	EDC grid website https://www.edc.com.kh/Annually_page/annuallyReport
42	Designated National Authority (DNA) in Cambodia Cambodia DNA Link
43	IFI Dataset of Default Grid Factors v.3.2 published in April 2022 Link
44	IGES List of Grid Emission Factors version 10.12 https://www.iges.or.jp/en/pub/list-grid-emission-factor/en
45	JCM website https://www.jcm.go.jp/kh-jp/methodologies/46/approved_pdf_file

46	Public data of the average salary level in Cambodia https://www.averagesalarysurvey.com/zh/salary/cambodia
47	'Corporate Finance: Theory and Practice', 2nd edition, by Aswath Damodaran'
48	IMF, World Economic Outlook Database IMF, World Economic Outlook Database
49	Financial Technical Due Diligence Report - TDD dated 19 November 2020 prepared by AFRY (Thailand) Limited ("AFRY")
50	JMR Report dated from November 2022 to December 2023.
51	IRR Sheet
52	Project Administration Manual of the National Solar Park
53	Annual Environmental and Social Safeguards Monitoring Report (ASMR) for the period January 2022 to December 2022 dated 10/01/2023
54	Corporate Tax Rate https://taxsummaries.pwc.com/cambodia/corporate/taxes-on-corporate-income
55	Tax holiday – incentives and schemes published by Council for the development of Cambodia https://cdc.gov.kh/incentives-and-schemes/
56	Waste and Waste Water Management Plan for the project

APPENDIX 3: JOINT VALIDATION AND VERIFICATION FINDINGS

Table 1: Clarification Requests (CLs) from this Joint Validation and Verification:

CL ID	01	**Date :** 15/05/2024
Description of CL		
Proof shall be provided for Construction of project and Commercial Operation as stated in section 1.1.		
Project participant response		**Date :** 03/10/2024
1. Engineering, Procurement and Construction Contract enclosed 2. Energization and Grid Synchronization Certificate enclosed		

Documentation provided by project participant	
EPC contract Grid synchronization certificate	
VVB assessment	**Date:** 02/11/2024
PP has signed EPC Contract with China CACS Engineering Corporation on 13/08/2021 and the same was verified by the VVB by checking the EPC Contract provided by PP. The commissioning of the project was verified by checking the Energization and Commissioning Test report provided by EDC dated 11/11/2022. Thus, CL 01 is closed.	

CL ID	02	Date : 15/05/2024
Description of CL		
PP shall submit the Power Purchase Agreement with Electricite Du Cambodge (EDC) for verification.		
Project participant response		**Date :** 03/10/2024
1. PPA enclosed		
Documentation provided by project participant		
Power Purchase Agreement		
VVB assessment		**Date:** 02/11/2024
PPA provided by PP was checked and found to be acceptable. Thus CL 02 is closed.		

CL ID	03	Date : 15/05/2024
Description of CL		
PP shall submit the all the necessary permits/clearances/consents obtained for the implementation of the project activity. Further, PP shall provide the documents pertaining to Ownership of the project activity as stated in section 1.8 of the JPDMR.		
Project participant response		**Date :** 03/10/2024
1. Bid award certificate enclosed 2. Land lease agreement enclosed 3. PPA enclosed 4. EIA license enclosed 5. EPC contract enclosed Further, section 1.8 of the JPDMR has been updated accordingly		
Documentation provided by project participant		
Bid award certificate Land lease agreement PPA EIA license EPC contract		
VVB assessment		**Date:** 02/11/2024
Assessment team checked the Letter of Award, Land lease agreement, PPA, EIA License and EPC Contract provided by PP and found to be acceptable. Thus CL 03 is closed.		

CL ID	04	Date : 15/05/2024
Description of CL		
As the project activity is bid-based, bid related documents shall be provided for verification.		
Project participant response		**Date :** 03/10/2024
1. Bid award certificate enclosed		
Documentation provided by project participant		
Bid award certificate		
VVB assessment		**Date:** 02/11/2024

Assessment team checked the Letter of Award provided by PP and found to be acceptable. Thus CL 04 is closed.

CL ID	05	Date : 15/05/2024
Description of CL		
In section 1.1 of the JPDMR, it is stated as the project activity serves as Phase 1 of the larger National Solar Park with a combined capacity of 100 MWac. However, the project is activity is only of 60 MW capacity. Further clarification on this shall be provided.		
Project participant response		**Date :** 03/10/2024
The National Solar Park Project involves the construction of two phases totaling 100- megawatt (MW) capacity in Kampong Chhnang province. The solar power plants were tendered to IPPs in two phases. The first phase for 60 MW. The second phase to procure 40 MW, for a total of 100 MW. The first phase of 60 MW was awarded to Prime Road Alternative Co., Ltd through a bid. Superfluous details have been removed and section 1.1 of the JPDMR has been updated.		
Documentation provided by project participant		
Bid award certificate Project Administration Manual of the National Solar Park		
VVB assessment		**Date:** 02/11/2024
Revised PDMR and the supporting documents provided by PP was checked by the assessment team and found to be acceptable. Thus CL 05 is closed.		

CL ID	06	Date : 15/05/2024
Description of CL		
Project proponent declaration for the emission reductions claimed under this project activity will not be double counted under any other GHG program to be submitted to VVB.		
Project participant response		**Date :** 03/10/2024
Developer’s declaration enclosed		
Documentation provided by project participant		
Developer’s Declaration Letter		
VVB assessment		**Date:** 02/11/2024
Declaration of PP on double counting was checked by assessment team and found to be acceptable. Thus CL 06 is closed.		

CL ID	07	Date : 15/05/2024
Description of CL		
PP shall provide a justification on why the degradation factor of solar panels is not taken into consideration while estimating the annual generation as annualized generation is equivalent throughout the crediting period.		
Project participant response		**Date :** 03/10/2024
Degradation factor has been now considered and subsequently the annual generation and emission reduction numbers have been revised and updated.		
Documentation provided by project participant		
Technical Due Diligence Report ER sheet		
VVB assessment		**Date:** 02/11/2024
Degradation factor considered from TDD report and the revised ER sheet have been checked by the assessment team and found to be acceptable. Thus CL 07 is closed.		

CL ID	08	**Date :** 15/05/2024
Description of CL		
PP shall provide the equipment supplier's specification to confirm the age and lifetime of the project activity.		
Project participant response		**Date :** 03/10/2024
Different equipment's have different design life. The overall lifetime of the project is 20 years and is in line with the PPA and Technical Due Diligence Report		
Documentation provided by project participant		
Technical Due Diligence Report		
VVB assessment		**Date:** 02/11/2024
Technical Due Diligence Report and PPA was provided by PP for the age and lifetime of project activity which was checked by the assessment team and found to be acceptable. Thus CL 08 is closed.		

CL ID	09	**Date :** 15/05/2024
Description of CL		
PP shall provide the proof of employment and trainings for the current monitoring period.		
Project participant response		**Date :** 03/10/2024
1. Employment declaration for the current monitoring period enclosed 2. O&M & Environment and Safety related training records & photographs enclosed		
Documentation provided by project participant		
Employment declaration by the project proponent Training records		
VVB assessment		**Date:** 02/11/2024
Employment declaration and Training records provided by PP was checked and found acceptable. Thus CL 09 is closed.		

CL ID	10	**Date :** 15/05/2024
Description of CL		
PP shall provide the following for the current monitoring period: 1. Joint meter reading reports 2. Electricity Invoices 3. Calibration Records		
Project participant response		**Date :** 03/10/2024
1. JMR for the current monitoring period enclosed 2. Electricity invoices for the current monitoring period enclosed 3. Calibration certificate enclosed		
Documentation provided by project participant		
JMR Electricity invoices Calibration certificate		
VVB assessment		**Date:** 02/11/2024
JMR, electricity invoices and calibration records were checked by the assessment team and found that there is a delay in calibration. PP has applied calibration error in the calculation which is acceptable to the assessment team. Thus CL 10 is closed.		

CL ID	11	**Date :** 15/05/2024

Description of CL	
PP shall provide the below listed documents: a. Detailed Project Report (DPR) b. Loan Sanction Letter c. EPC Contract d. O & M Contract e. Common Terms Agreement f. Board Resolution/ Note on investment g. Land Lease Agreement h. Proof for Insurance and Overheads i. Basis for Tax rate considered	
Project participant response	**Date :** 03/10/2024
1. Technical Due Diligence Report enclosed which has information on insurance, overheads and tax rate considered 2. Common Terms Agreement enclosed; it serves as the proof of loan sanctioned/note on investment 3. EPC contract enclosed 4. O&M contract enclosed 5. Land lease agreement enclosed	
Documentation provided by project participant	
Technical Due Diligence Report Common Terms Agreement EPC contract O&M contract Land lease agreement	
VVB assessment	**Date:** 02/11/2024
The above stated documents provided by PP were checked and found to be ok. Thus CL 11 is closed.	

CL ID	12	Date : 15/05/2024
Description of CL		
PP shall submit the proof of Local Stake Consultation- a. Invitation, MOM, Interview list, Feedback from stakeholders, Photographs, etc. b. ESIA Report		
Project participant response		**Date :** 03/10/2024
Initial Environmental Examination (IEE) report, chapter 5, provides all the relevant details of the village and provincial level consultations held. Subsequent LSCs were conducted, the information of the same is captured in the Annual Environmental and Social Safeguards Monitoring Report (ASMR).		
Documentation provided by project participant		
IEE report ASMR reports		
VVB assessment		**Date:** 02/11/2024
The above stated documents provided by PP were checked and found to be ok. Thus CL 12 is closed.		

CL ID	13	Date : 15/05/2024
Description of CL		
In section 2.1.4 of the JPDMR, it is stated that the key contacts for the GRC have been displayed at the project site... However, no such was displayed at project location. Clarification requested		
Project participant response		**Date :** 03/10/2024
The key contacts for the GRC were handed over to the stakeholders in person during the public consultation. Section 2.1.4 has been revised accordingly.		

Documentation provided by project participant	
VVB assessment	**Date:** 02/11/2024
The revised PDMR was checked for the stated revision which is found to be correct. Thus CL 13 is closed.	

CL ID	14	Date : 15/05/2024
Description of CL		
The policies of PP as stated in section 2.3.1 shall be provided for verification.		
Project participant response		**Date :** 03/10/2024
1. Anti-sexual harassment policy enclosed 2. HR policy enclosed		
Documentation provided by project participant		
Anti-sexual harassment policy HR policy enclosed		
VVB assessment		**Date:** 02/11/2024
The above stated documents provided by PP were checked and found to be ok. Thus CL 14 is closed.		

CL ID	15	Date : 15/05/2024
Description of CL		
In section 2.4 of the JPDMR, a. PP has stated that soil quality monitoring will be carried out every six months. Proof for the same shall be provided. b. PP states that the water used for panel cleaning will be recycled. How will this be done? Proof and details on the same shall be provided. c. PP shall provide the proof of all the mitigation or preventive measures taken in this context.		
Project participant response		**Date :** 03/10/2024
a. Soil quality monitoring and testing was carried out for six months during the construction period, section 2.4 has been corrected now. b. Water used for panel cleaning is not being recycled, this piece of information has been retracted.		
Documentation provided by project participant		
IEE report ASMR report		
VVB assessment		**Date:** 02/11/2024
The above stated documents provided by PP were checked and found to be ok. Thus CL 15 is closed.		

Table 2: Corrective Action Requests (CARs) from this Joint Validation and Verification:

CAR ID	01	Date : 15/05/2024
Description of CAR		
The title of the template in page 4 is incorrectly mentioned. The same shall be corrected by PP.		
Project participant response		**Date :** 03/10/2024

Title of the template in page 4 is corrected	
Documentation provided by project participant	
VVB assessment	**Date:** 02/11/2024
The title of the template revised in the JPDMR was checked and found to be correct. Thus CAR 01 is closed.	

CAR ID	02	**Date :** 15/05/2024
Description of CAR		
During the site audit, the project activity was found to be bid-based project. The details on the same shall be incorporated in section 1.1 of the JPDMR. Correction requested.		
Project participant response		**Date :** 03/10/2024
Bid related detail is now updated in the section 1.1 of the JPDMR.		
Documentation provided by project participant		
Tender Award Letter enclosed		
VVB assessment		**Date:** 02/11/2024
The bid related details included in the updated JPDMR was checked and found to be correct. Thus CAR 02 is closed.		

CAR ID	03	**Date :** 15/05/2024
Description of CAR		
Under section 1.8 of the JPDMR, PP shall clearly state under which point of paragraph 3.7.1 (points 1 to 7) of the VCS standard the ownership of the project activity arises and evidence of the same shall be submitted to the VVB.		
Project participant response		**Date :** 03/10/2024
The ownership of the project activity is in-line with the VCS Standard, v4.7, subsection 3.7.1 requirement 3. This is updated in the JPDMR section 1.8 and evidence of the same will be submitted.		
Documentation provided by project participant		
Tender Award certificate EPC contract Land lease agreement EIA license PPA		
VVB assessment		**Date:** 02/11/2024
Above stated documents were checked and found that the ownership evidence included in the updated JPDMR is correct. Thus CAR 03 is closed.		

CAR ID	04	**Date :** 15/05/2024
Description of CAR		
In section 1.12 of the JPDMR, a. PP shall provide the Plant Load Factor (PLF) details and clarify how the PLF meets the requirement of CDM EB Guidelines (EB 48 annex 11) for the reporting and validation of PLF. b. PP shall include the details of number of solar panels. Also, include the details on number of meters, their arrangement and their location.		

Project participant response	**Date :** 03/10/2024
a. The Plant Load Factor is provided by a third-party engineering company, "AFRY (Thailand) Limited", and is line with the CDM EB 48 annex 11. The proof of the same is in the Technical Due Diligence Report. b. Details regarding no. of solar panels and meters are now updated in the section 1.12 of the JPDMR	
Documentation provided by project participant	
Technical Due Diligence Report	
VVB assessment	**Date:** 02/11/2024
PLF is calculated by the generation data (P90) given in the TDD report prepared by third party- *AFRY (Thailand) Limited ("AFRY"), which is acceptable to the assessment team. Thus,* the considered PLF meets the requirement of CDM EB Guidelines (EB 48 annex 11) for the reporting and validation of PLF. Further, the updated technical details in the revised JPDMR were checked and found to be ok. Thus CAR 04 is closed.	

CAR ID	05	**Date :** 15/05/2024
Description of CAR		
The geographical coordinates provided for the project location in section 1.13 of the JPDMR do not correspond to the actual project activity. Correction requested.		
Project participant response		**Date :** 03/10/2024
The geo-coordinates are now updated in section 1.13. However, this does not reflect in the google maps, please cross verify on google earth.		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
Revised Co-ordinates provided in the updated JPDMR is correct. The same was verified from Google Earth Pro/Maps. Thus CAR 05 is closed.		

CAR ID	06	**Date :** 15/05/2024
Description of CAR		
Section 1.18.1 of the JPDMR doesn't address all the requirements of template. PP shall check and include the required details in section 1.18.1 of the JPDMR. Correction requested.		
Project participant response		**Date :** 03/10/2024
Section 1.18.1 of the JPDMR is now updated with details on how the project results in SD, it's contributions to the host countries SD priorities.		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 06 is closed.		

CAR ID	07	**Date :** 15/05/2024
Description of CAR		
For all the acronyms used in JPDMR, the full-forms shall be provided at the first instance.		
Project participant response		**Date :** 03/10/2024
All the acronyms are now provided with full-forms at the first instance		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024

The revised JPDMR has full-forms of all the acronyms used at the first instance. Thus, CAR 07 is closed.

CAR ID	08	Date : 15/05/2024
Description of CAR		
In the IRR sheet, PP shall include the source for all the input parameters considered in IRR calculations.		
Project participant response		**Date :** 03/10/2024
The sources off input parameters in the IRR sheet is Technical Due Diligence Report, and this source is mentioned now and will be provided for verification		
Documentation provided by project participant		
Technical Due Diligence Report IRR sheet		
VVB assessment		**Date:** 02/11/2024
The source document for all the input parameters, i.e., TDD is updated in the JPDMR and IRR sheet, which is found to be acceptable to the assessment team. Thus CAR 08 is closed.		

CAR ID	09	Date : 15/05/2024
Description of CAR		
Project proponent is required to complete section 2.1.5 of the JPDMR since the public commenting period for the project has been completed. Correction requested.		
Project participant response		**Date :** 03/10/2024
Section 2.1.5 is now updated.		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 09 is closed.		

CAR ID	10	Date : 15/05/2024
Description of CAR		
Under section 3.3 of the JPDMR, the source for the project case shall be included.		
Project participant response		**Date :** 03/10/2024
Source of the project boundary diagram in section 3.3 is provided and the link to the same source is provided as a foot note.		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 10 is closed.		

CAR ID	11	Date : 15/05/2024
Description of CAR		
PP shall include the investment decision date for the project activity in section 3.5.2 of the JPDMR. Please provide the proof for the same.		
Project participant response		**Date :** 03/10/2024
In section 3.5.2, investment decision date is updated now and EPC contract will be provided to the VVB as proof of the same		
Documentation provided by project participant		

EPC Contract	
VVB assessment	**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the investment decision date and found to be ok. Thus CAR 11 is closed.	

CAR ID	12	**Date :** 15/05/2024
Description of CAR		
In section 3.5.2 of the JPDMR, Under Step 1, the sub-steps and their outcomes shall be documented. Under Step 2, the sub-step 2a shall be included.		
Project participant response		**Date :** 03/10/2024
Section 3.5.2 of the JPDMR is now updated with: a. Sub-step 1a and it's outcomes b. Sub-step 1b and it's outcomes c. Sub-step 2a is included now		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 12 is closed.		

CAR ID	13	**Date :** 15/05/2024
Description of CAR		
In section 3.5.2 of JPDMR, under 'Common Practice Analysis', a. PP shall include the basis for the start date considered for common practice analysis and shall provide the proof for the same. b. Only the number of projects considered have been given. Hence, PP shall provide the details on which are those projects considered for common practice analysis. Correction requested.		
Project participant response		**Date :** 03/10/2024
a. In section 3.5.2, the common practice analysis has been carried out as per CDM methodological tool for "Common Practice". Therefore the start date is as per CDM glossary which states that, " *For a CDM project activity (non-A/R) or CPA (non-A/R), the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system), for the CDM project activity or CPA. Where a contract is signed for such expenditures (e.g. for procurement of a wind turbine), it is the date on which the contract is signed*." Therefore, the date on which EPC contract was signed i.e 13th August 2021 has been considered as the start date for assessing common practice and the same has been now updated in the JPDMR. b. The details on the public sources from which the projects have been considered for common practice analysis is provided as foot notes. Further, VVB will be provided with the CPA sheet.		
Documentation provided by project participant		
EPC contract CPA sheet		
VVB assessment		**Date:** 02/11/2024

The updated JPDMR, CPA sheet and EPC Contract was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 13 is closed.

CAR ID	14	**Date :** 15/05/2024
Description of CAR		
In section 5.2 of the JPDMR, the equation given is not as per the applied methodology. Correction requested.		
Project participant response		**Date :** 03/10/2024
In section 5.2, the equation to account for Project Emission is now updated.		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the equations on project emissions and found to be ok. Thus CAR 14 is closed.		

CAR ID	15	**Date :** 15/05/2024
Description of CAR		
In section 5.4 of the JPDMR, PP shall include the methodology reference for the equation considered. Correction requested.		
Project participant response		**Date :** 03/10/2024
Documentation provided by project participant		
In section 5.4 of the JPDMR, the methodology reference is now included and the formula has been updated accordingly.		
VVB assessment		**Date:**02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 15 is closed.		

CAR ID	16	**Date :** 15/05/2024
Description of CAR		
In section 6.1 of JPDMR, PP shall include the document published date and reference link for the source of data.		
Project participant response		**Date :** 03/10/2024
Section 6.1 of the JPDMR is now updated with the published date and reference link to the source		
Documentation provided by project participant		
VVB assessment		**Date:** 02/11/2024
The updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Thus CAR 16 is closed.		

CAR ID	17	**Date :** 15/05/2024
Description of CAR		
In section 6.2 of JPDMR, 1. ‘Description of measurement methods and procedures applied’ section shall be filled as per the requirements of template. 2. Calibration frequency is considered as 5 years. However, it is not clear based on what standards/ protocol this was considered. The reference for the same shall be included.		
Project participant response		**Date :** 03/10/2024

Section 6.2 of the JPDMR is now updated with: 1. Measurement process and metering equipment accuracy details 2. The calibration frequency has been revised as per the frequency mentioned in the calibration certificate, where the document specifies that frequency is a, "yearly calibration work."
Documentation provided by project participant
Meter calibration Certificate
VVB assessment — **Date:** 02/11/2024
Section 6.2 of updated JPDMR was checked by the assessment team for the relevant revisions and found to be ok. Calibration records were checked found that there is a delay in calibration during the current monitoring period. PP has applied calibration error in the calculation which is acceptable to the assessment team. Thus CAR 17 is closed.

CAR ID	18	Date : 15/05/2024
Description of CAR		
1. In section 6.3 of JPDMR, the following information as per the requirements of template are found missing: a. The methods used for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the processes used for calibrating monitoring equipment. b. Organizational structure c. Procedures for internal auditing d. Procedures for handling non-conformances with the validated monitoring plan. e. Include line diagrams to display the GHG data collection and management system, Where appropriate. 2. Further, the monitoring practices followed at site is not matching with the one provided in section 6.3 of the JPDMR. Correction requested.		
Project participant response		**Date :** 03/10/2024
The section 6.3 has now been updated as per the JPDMR specifications and is in line with the approach mentioned in the PPA as well as being followed at the site.		
Documentation provided by project participant		
PPA		
VVB assessment		**Date:** 02/11/2024
Section 6.2 of updated JPDMR was checked by the assessment team for the corrections in monitoring plan and found to be ok. Thus CAR 18 is closed.		

CAR ID	19	Date : 15/05/2024
Description of CAR		
In section 7.5 of the JPDMR, 1. Please include the relevant equations in quantifying the GHG emission reductions. 2. The instructions from the JPDMR template are retained. Kindly check and remove the same.		
Project participant response		**Date :** 03/10/2024
Section 7.5 of the JPDMR is now updated with: 1. Equation of GHG emission reductions 2. Instructions from the JPDMR template is removed		

Documentation provided by project participant	
VVB assessment	**Date:** 02/11/2024
Section 7.5 of the JPDMR is updated with the equations utilized in quantifying emission reductions. Additionally, the instructions from the template have been removed. Thus, CAR 19 is closed.	

CAR ID	20	**Date :** 15/05/2024
Description of CAR		
The title of the project mentioned in ER sheet is not consistent with JPDMR/registry. Also, project reference ID shall be included. Correction required.		
Project participant response		**Date :** 03/10/2024
The title of the project in the ER sheet has revised as per the name mentioned in the JPDMR/registry and the project reference ID has been included.		
Documentation provided by project participant		
Estimated & Actual ER calculation sheets		
VVB assessment		**Date:** 02/11/2024
The title of the project activity provided in the updated ER sheets is consistent with JPDMR/ registry. Thus CAR 20 is closed.		

Table 3: Forward Action Requests (FARs) from this Joint Validation and Verification:

~~**FAR ID**~~		~~**Date :**~~
~~**Description of CAR**~~		
~~**Project participant response**~~		~~**Date :**~~
~~**Documentation provided by project participant**~~		
~~**VVB assessment**~~		~~**Date:**~~

APPENDIX 4: ABBREVIATIONS

4KES	4K Earth Science Private Limited
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CP	Crediting Period
EB	Executive Board
EDC	Electricite Du Cambodge
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse Gases

IPCC	Intergovernmental Panel for Climate Change
ISO	International Organization for Standardization
JPDMR	Joint Project Description and Monitoring Report
LE	Local Expert
MP	Monitoring Period
MR	Monitoring Report
MW	Mega Watt
MWh	MegaWatt hour
NA	Not Applicable
PD	Project Description
PLF	Plant Load Factor
PP	Project Proponent
PPA	Power Purchase Agreement
QA/QC	Quality Assurance/Quality Control
tCO_2	Tonnes of Carbon Dioxide
TDD	Technical Due Diligence Report
TE	Technical Expert
TL	Team Leader
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCSA	Verified Carbon Standard Association
VCU	Verified Carbon Unit

APPENDIX 5: TEAM COMPETENCE

Certificate of Competence							
Name ☐ Mr. ☒ Ms.	**Swati S Acharya**						
Qualification Procedure	*Fulfils the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.*						
Appointed to work as:							
	Validator/ Verifier	**Team Leader**	**Trainee**	**Technical Expert**	**Technical Reviewer**	**Financial Expert**	**Approver**
Appointed	Yes	Yes	No	Yes	No	No	No
Appointed Date	05-12-2023						
Authorized to work as Technical Expert for:							
Authorized Technical Area	**Sectoral Scope**		**TA Code**		**Technical Area within the scope**		
	Energy industries (renewable - / non-renewable sources)		1.2		Renewables		

	Transportation	7.1	Transport
Authorized to work as Local Expert for:			
Country/Countries	India		
Compliance check by:		Anand S R	

Certificate of Competence							
Name	☒ Mr. ☐ Ms.	**Sriram S**					
Qualification Procedure		*Fulfils the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.*					
Appointed to work as:							
		Validator/ Verifier	**Team Leader**	**Trainee**	**Technical Expert**	**Technical Reviewer**	**Financial Expert**	**Approver**
Appointed		No	No	Yes	No	No	No	No
Appointed Date		15-07-2023						
Authorized to work as Technical Expert for:							
Authorized Technical Area		**Sectoral Scope**	**TA Code**	**Technical Area within the scope**			
		-	-	-			
Authorized to work as Local Expert for:							
Country/Countries		-					
Compliance check by:		Anand S R					

Certificate of Competence								
Name	☒ Mr. ☐ Ms.	**Ma Paa Puratchikkanal**						
Qualification Procedure		*Fulfils the requirement as per the appointment of personnel procedure of 4KES for Validation and Verification of CDM/VCS/GS/GCC/GHG Projects.*						
Appointed to work as:								
		Validator/ Verifier	**Team Leader**	**Trainee**	**Technical Expert**	**Technical Reviewer**	**Financial Expert**	**Approver**
Appointed		Yes	Yes	No	Yes	Yes	Yes	Yes
Appointed Date		15-07-2023						
Authorized to work as Technical Expert for:								
Authorized Technical Area		**Sectoral Scope**	**TA Code**	**Technical Area within the scope**				
		Energy industries (renewable - / non-renewable sources)	1.1	Thermal energy generation				
		Energy industries (renewable - / non-renewable sources)	1.2	Renewables				
		Energy demand	3.1	Energy demand				
		Construction	6.1	Construction				

	Waste handling and disposal	13.1	Solid waste and wastewater
	Waste handling and disposal	13.2	Manure
	Agriculture	15.1	Agriculture
Authorized to work as Local Expert for:			
Country/Countries	India and Sri Lanka		
Compliance check by:	Anand S. R.		